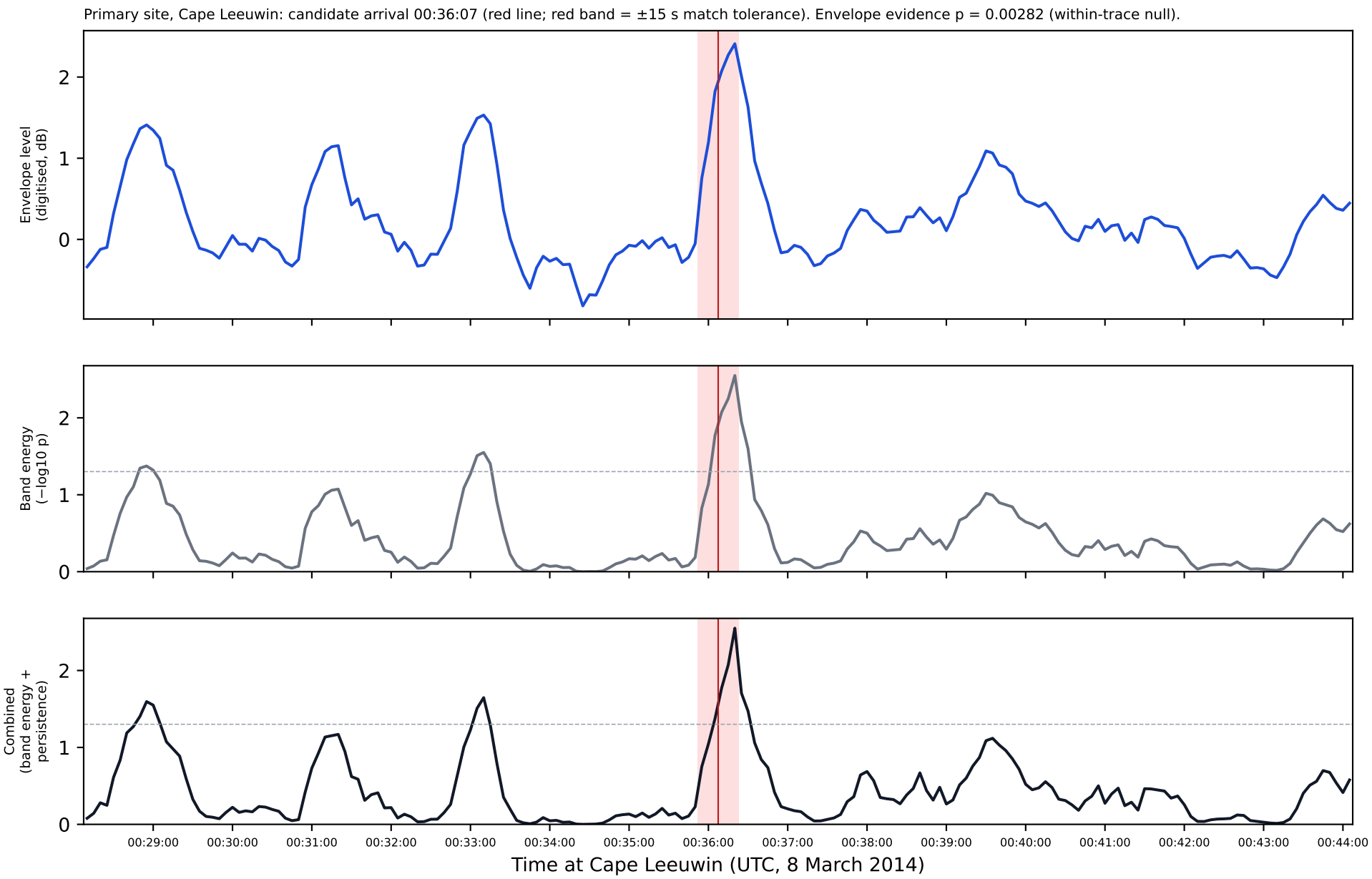
Ten strongest Cape Leeuwin and Diego Garcia South candidates with an implied source on the 7th arc

Rank	Station	Arrival	Bearing, published / corrected	Found by	Envelope p	Arc latitude (° S)	Implied impact	Steep dive	Catalogue label (4 Nov)
1	Cape Leeuwin	00:36:07	none	ours	0.003	any (no bearing)	00:19:38–00:22:23	yes	S4: EQ-EV081
2	Cape Leeuwin	00:47:04	343.2° / 338.7° (±2.0°)	Kadri and ours	0.006	15; 16 (outside 20-41)	00:19:38–00:26:56	yes	S3: EQ-ISC-606477921
3	Diego Garcia S.	01:10:15	none	ours	0.009	any (no bearing)	00:19:38–00:34:15	yes	
4	Cape Leeuwin	00:54:30	306.2° / 299.4° (±3.9°)	Kadri and ours	0.062	27	00:31:26–00:40:42	no	S4: EQ-ISC-606477922
5	Cape Leeuwin	00:51:50	320.4° / 315.5° (±3.9°)	Kadri and ours	0.096	23; 24	00:26:44–00:37:25	no	S4: EQ-ISC-606477922; S8: EQ-ISC-606477922
6	Cape Leeuwin	00:49:58	260.4° / 246.8° (±2.1°)	Kadri	0.121	38; 39; 40	00:19:38–00:26:28	yes	
7	Cape Leeuwin	00:55:07	314.1° / 308.5° (±3.9°)	Kadri	0.237	25; 26	00:31:10–00:41:05	no	
8	Cape Leeuwin	00:45:36	312.0° / 306.1° (±3.9°)	Kadri	0.273	25; 26	00:22:04–00:31:34	yes	
9	Cape Leeuwin	00:40:34	303.3° / 296.0° (±3.9°)	Kadri	0.332	28	00:19:38–00:26:50	yes	
10	Cape Leeuwin	00:41:43	310.0° / 303.8° (±3.9°)	Kadri	0.344	26; 27	00:19:38–00:27:55	yes	

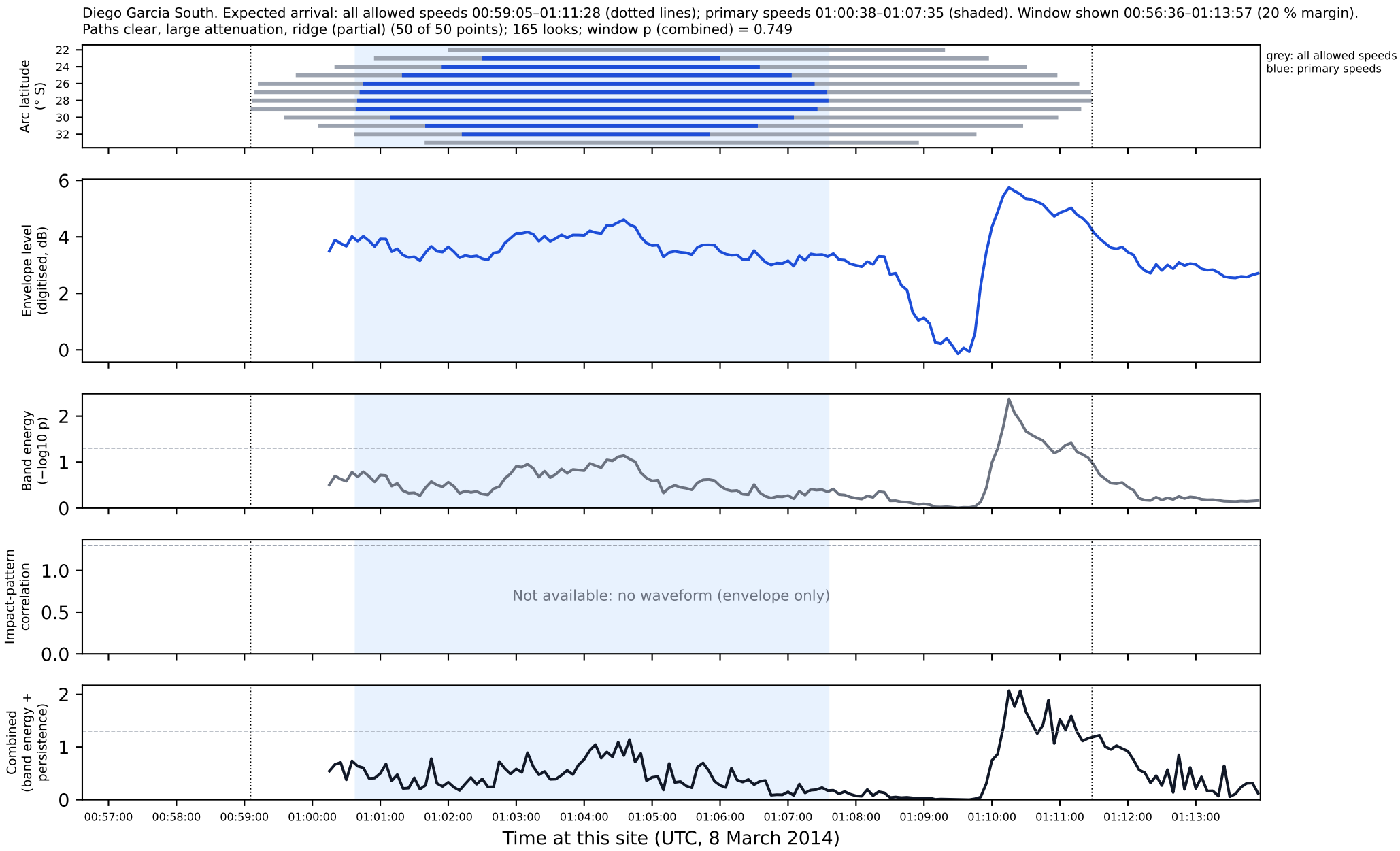
Ranking: envelope evidence at the station itself (band energy and persistence on the digitised Kadri (2024) Figure 9 envelope, smallest p within ±15 s; within-trace null). This is weak evidence: the p floor is about 1/355 and the 19 Kadri times are not corrected for multiple looks. Sources: Kadri Table 1; ours = combined-detector best looks on the envelopes and 4 Nov screening picks (merged within 30 s of a Kadri time, or 45 s of each other). Bearings: Kadri's published bearings corrected for the H01W array-position bias (calibration from 144 earthquakes, ruling B1; corrections -4.4 to -13.8°); tolerance ±1.645σ with σ = calibrated sector error and a 1.2° weak-signal term. Stronger candidates not shown: 00:39:53 (published 256.8°, corrected 243.0°; envelope p 0.093): bearing fits 40-43 S, but implied impact 23:54:52-00:12:08 is before 00:19:38; 00:49:41 (published 234.7°, corrected 221.7°; envelope p 0.121): corrected bearing meets no 7th-arc point; 00:38:29 (published 261.2°, corrected 247.7°; envelope p 0.214): bearing fits 38-43 S, but implied impact 23:52:52-00:14:59 is before 00:19:38; 00:39:02 (published 268.2°, corrected 255.3°; envelope p 0.251): bearing fits 36-40 S, but implied impact 00:03:32-00:18:54 is before 00:19:38. Implied impact: arc points 15-43° S within the bearing tolerance (no bearing: 20-41° S), sound speed 1.40 to 1.56 km/s, 5 % detour, inside 00:19:38 to 00:55:00. Steep dive: impact 00:19:38 to 00:29:44 within 25 NM. Catalogue labels are associations below 0.95 probability (label only, not removed).

Candidate 1: Cape Leeuwin 00:36:07 (ours; no bearing). Implied impact 00:19:38–00:22:23 UTC, arc latitude from timing only



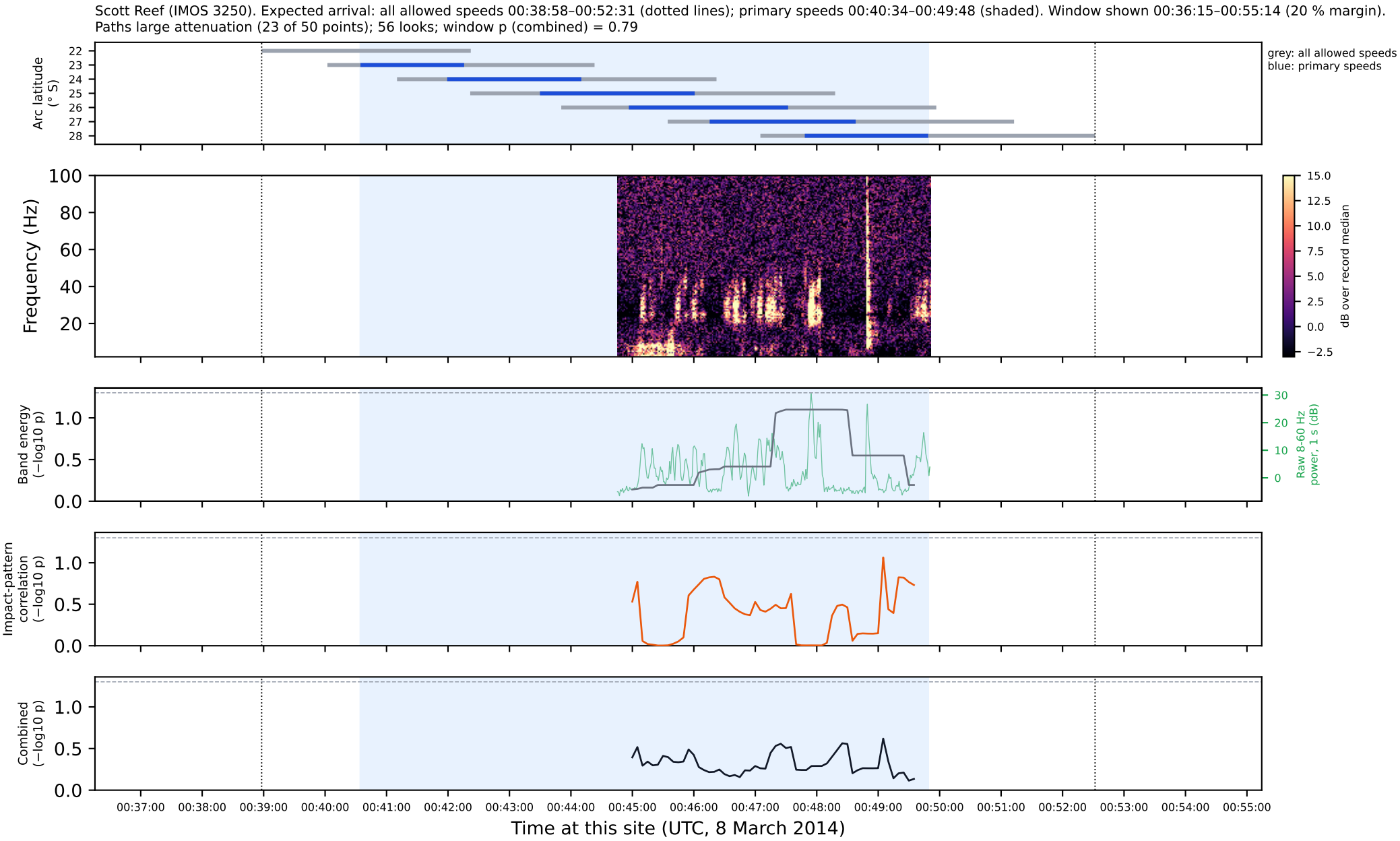
Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

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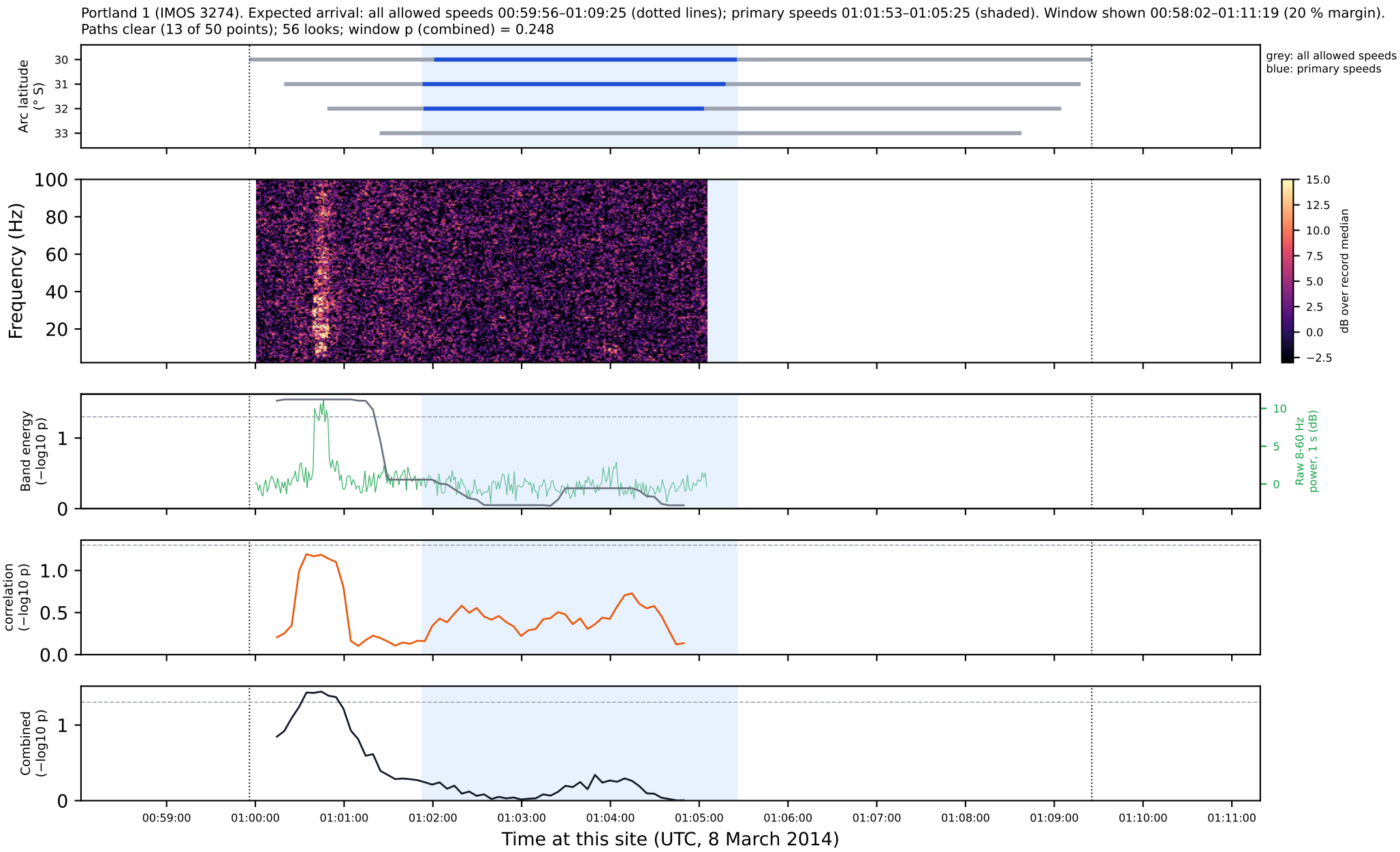
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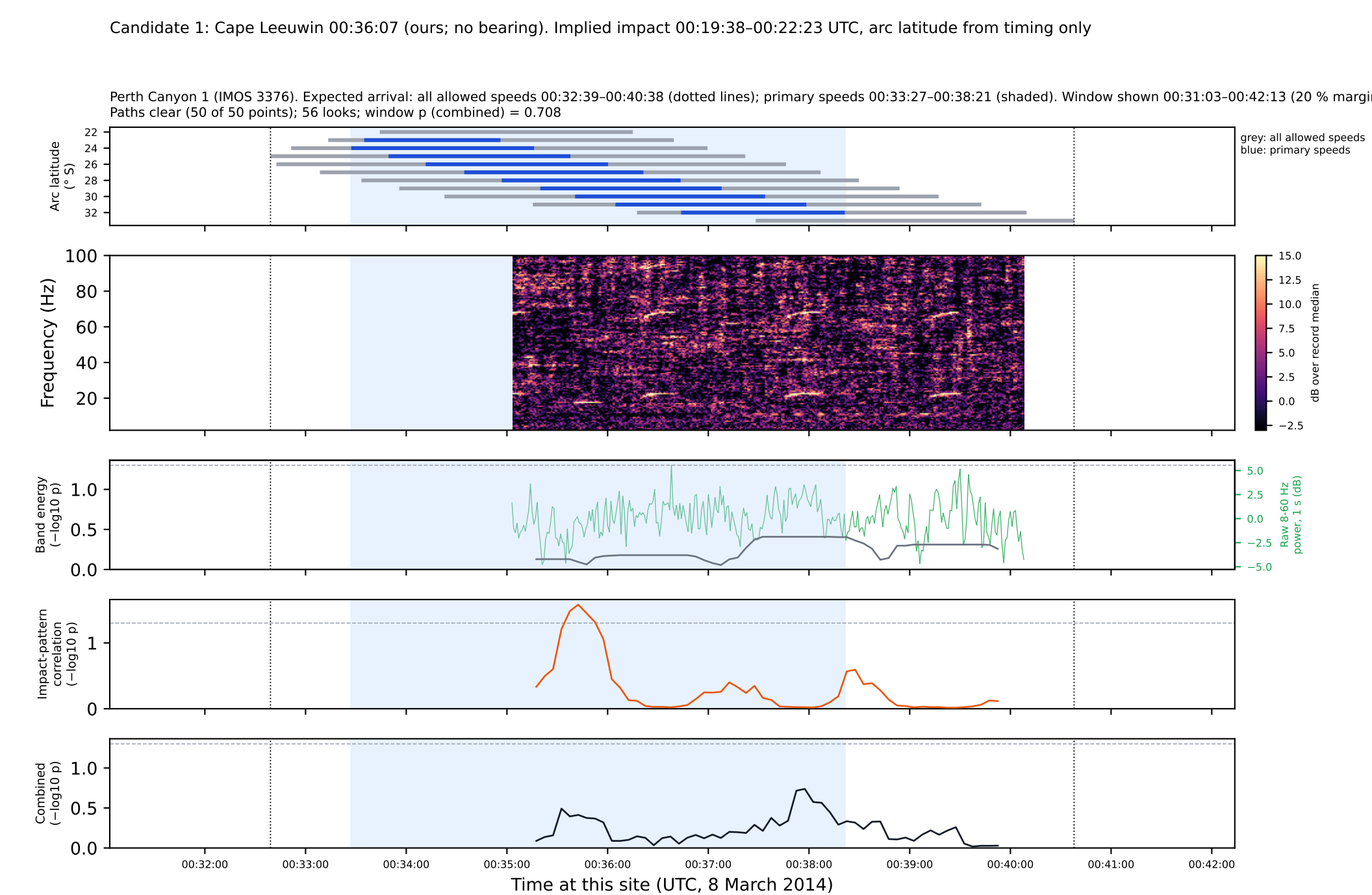


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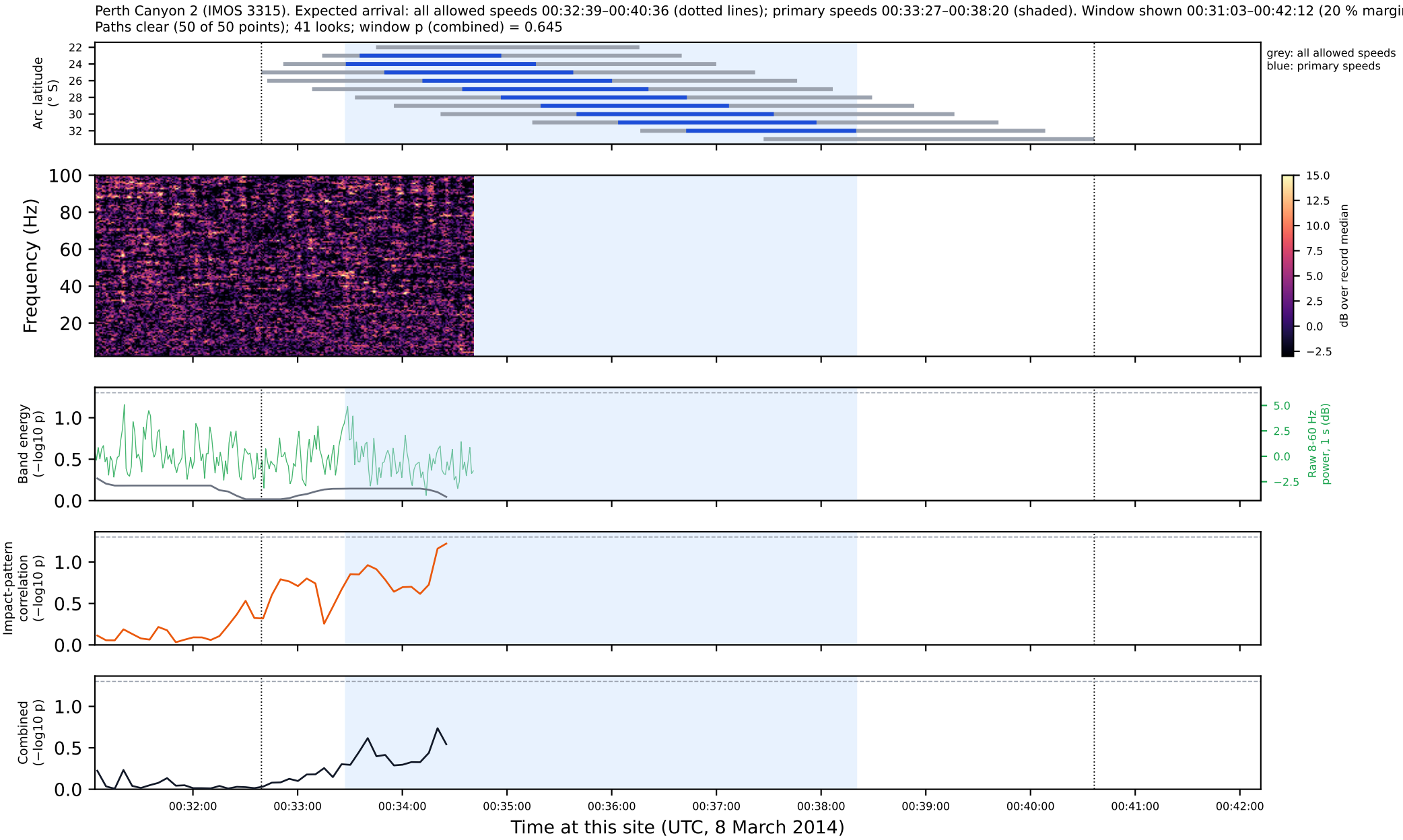


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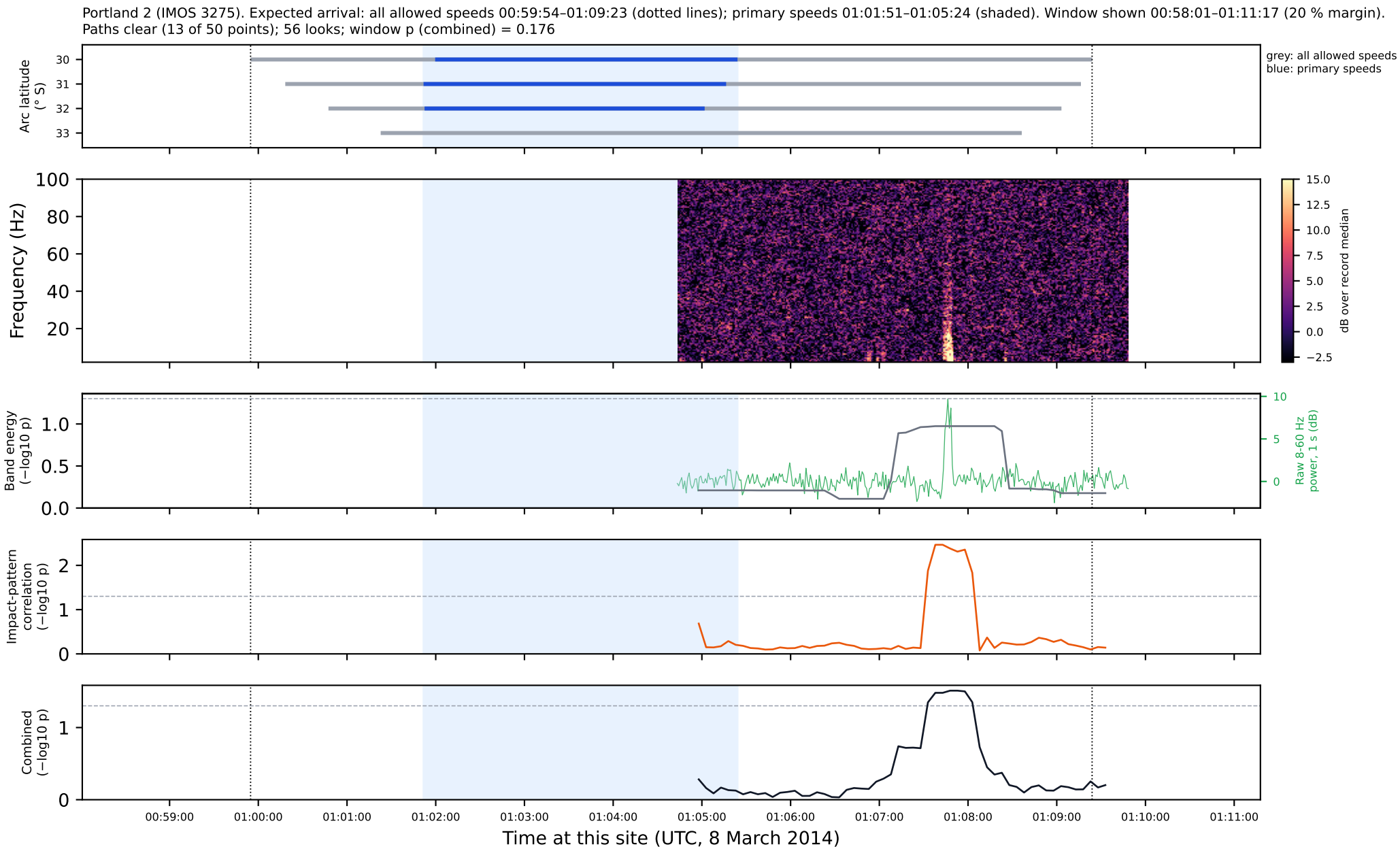
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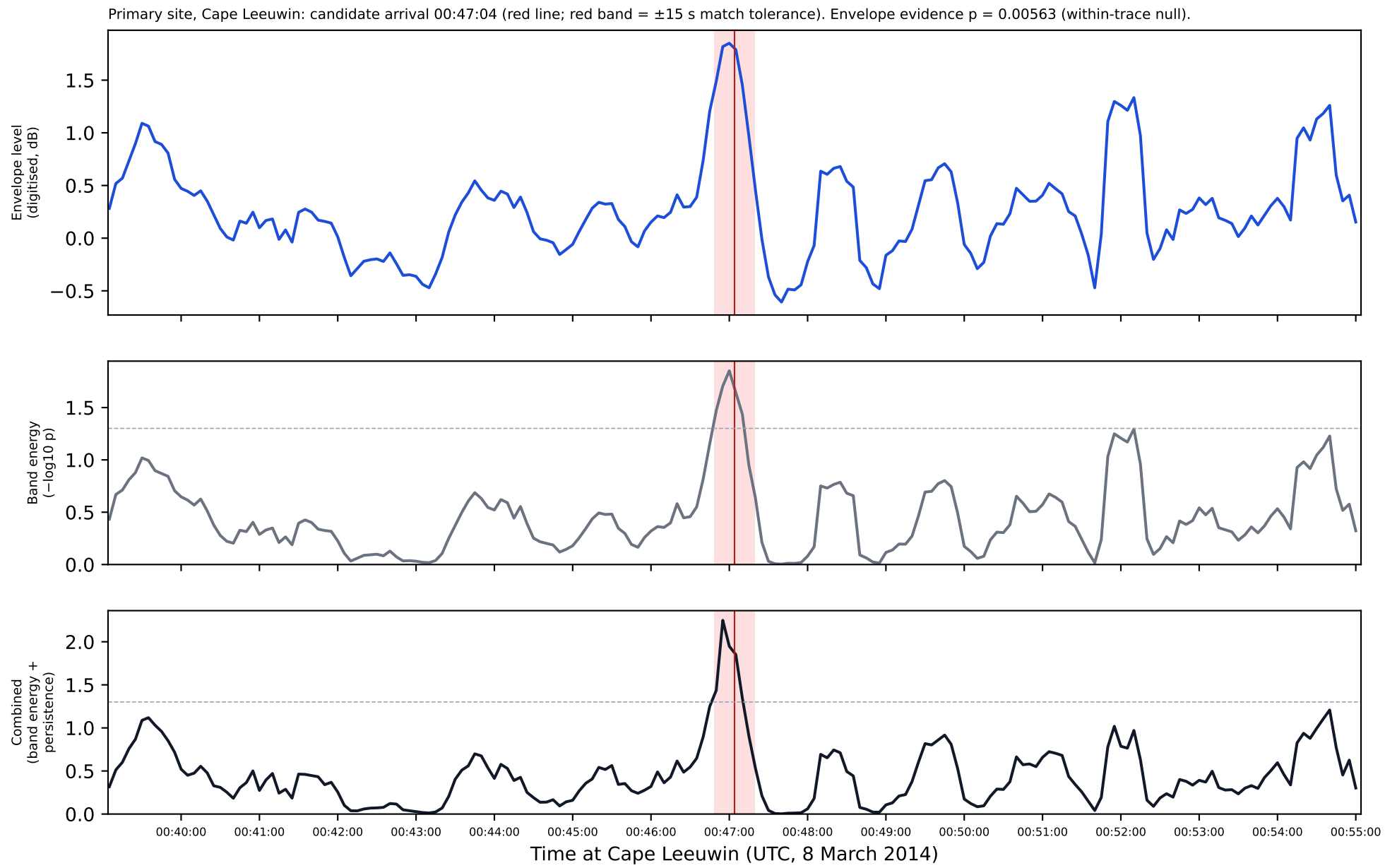
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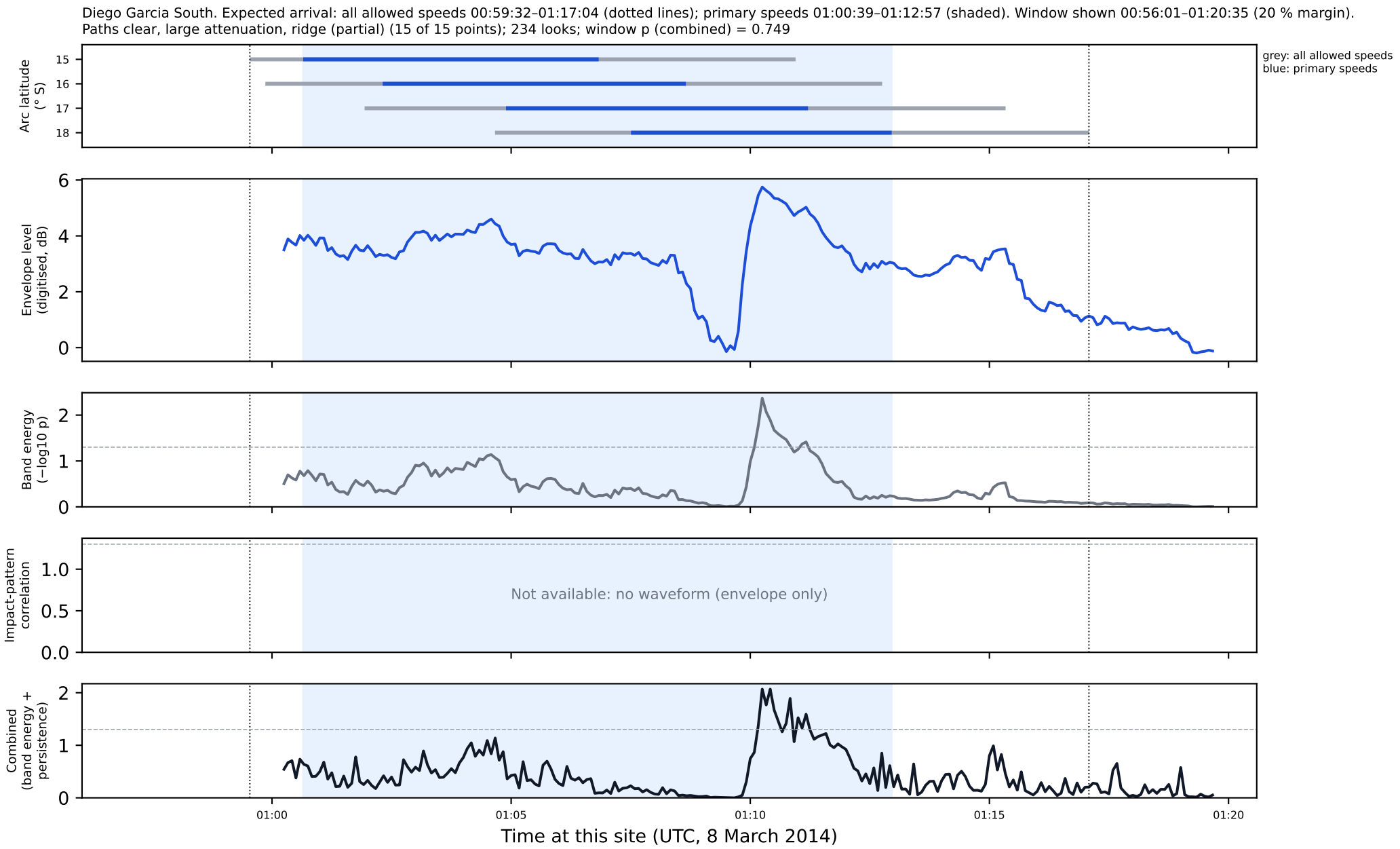
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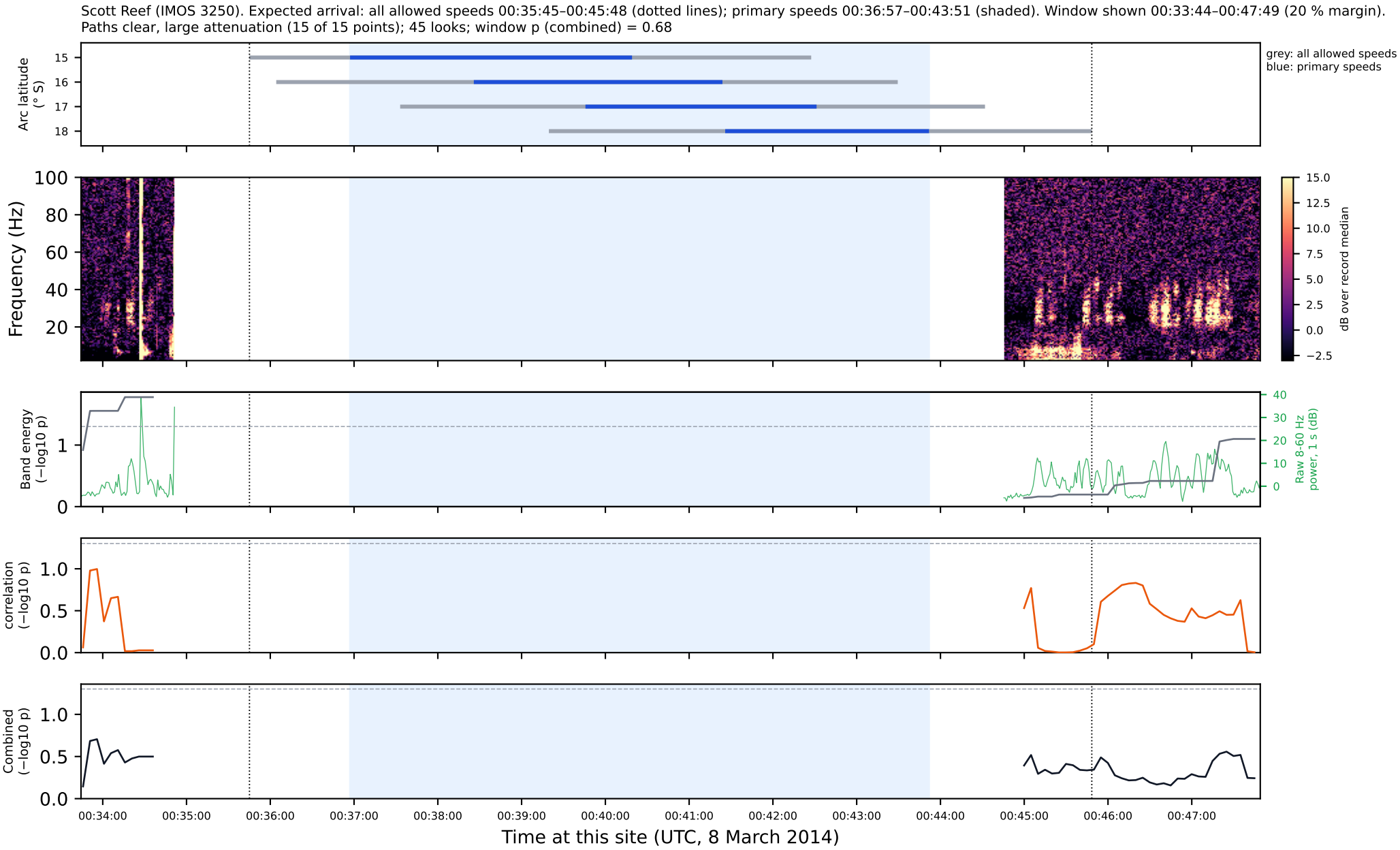
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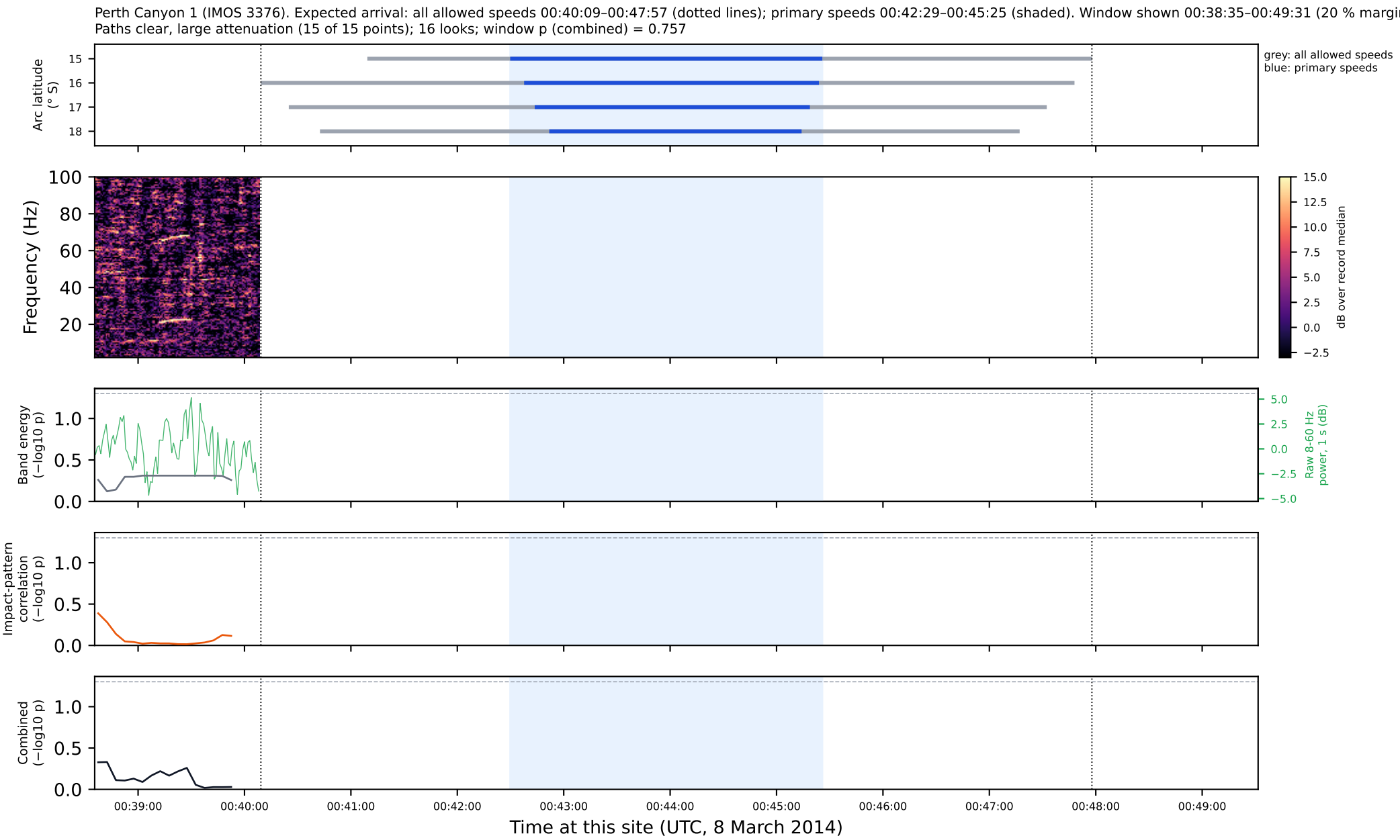
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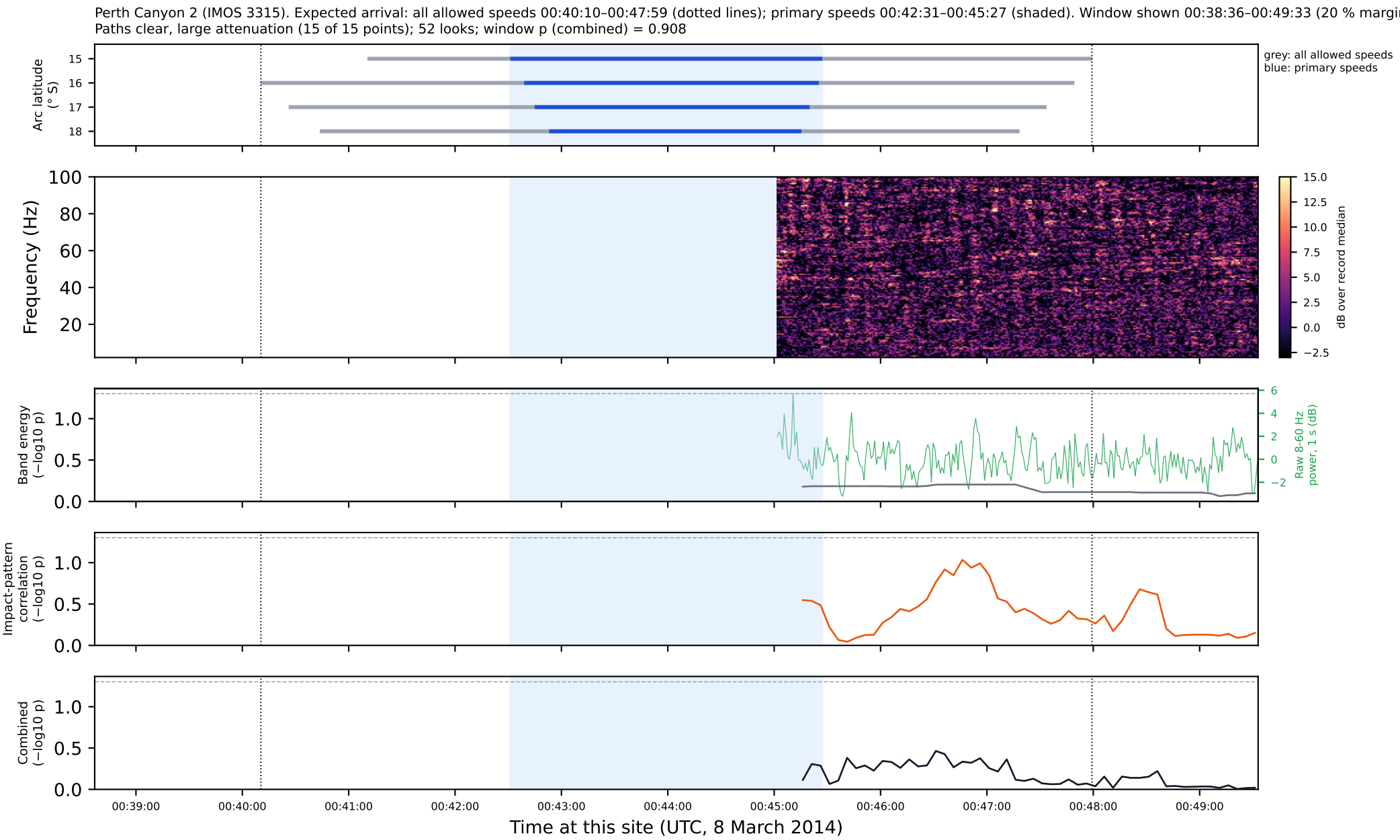
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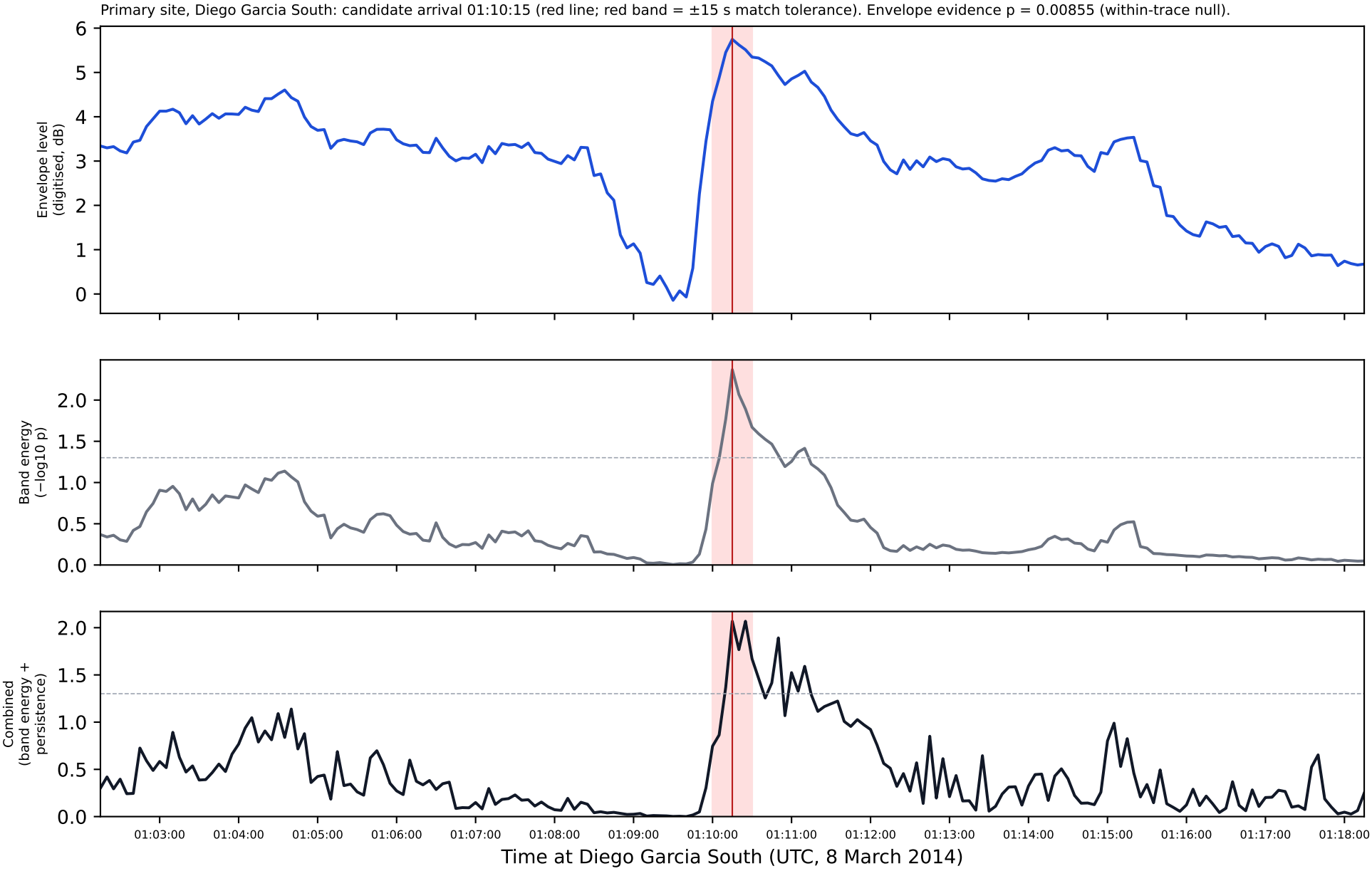
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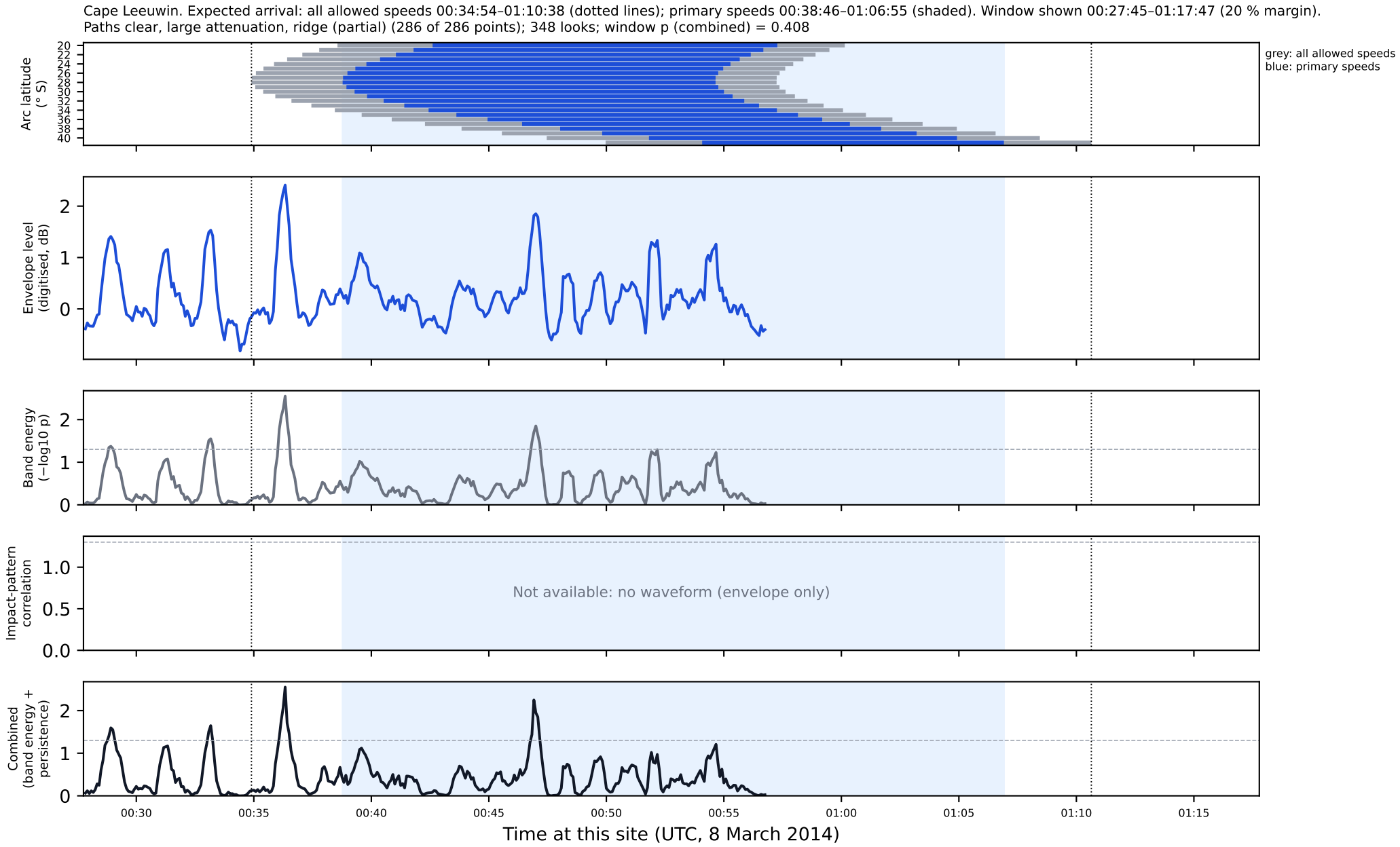
Window: earliest arrival at 1.56 km/s to latest at 1.40 km/s with 5 % path detour, over every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing), then widened by 20 % of its length at each end (outer dotted lines mark the unwidened span). Shaded: span for the primary sound speed 1.465 to 1.520 km/s with 2 % detour. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

Candidate 3: Diego Garcia South 01:10:15 (ours; no bearing). Implied impact 00:19:38–00:34:15 UTC, arc latitude from timing only



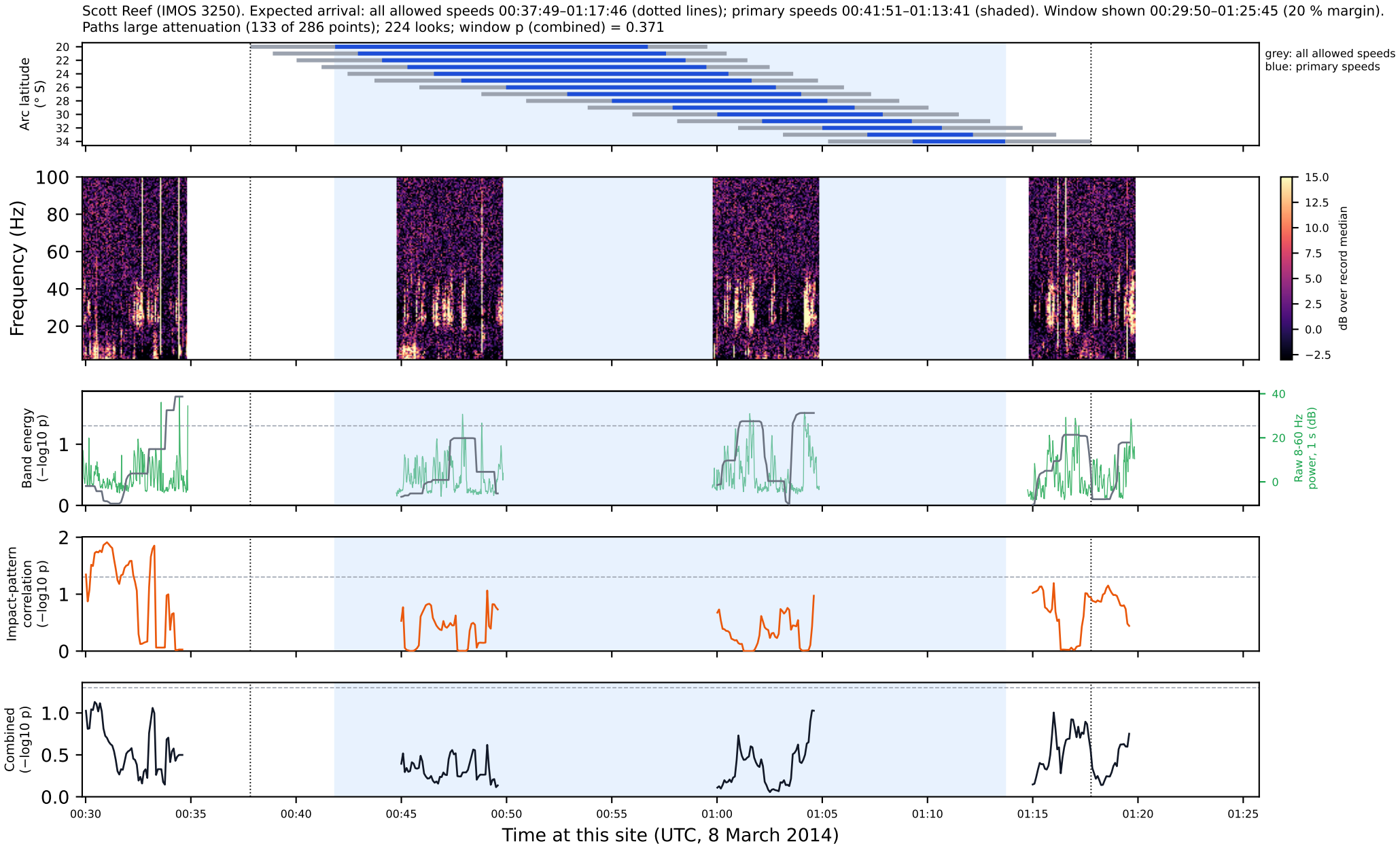
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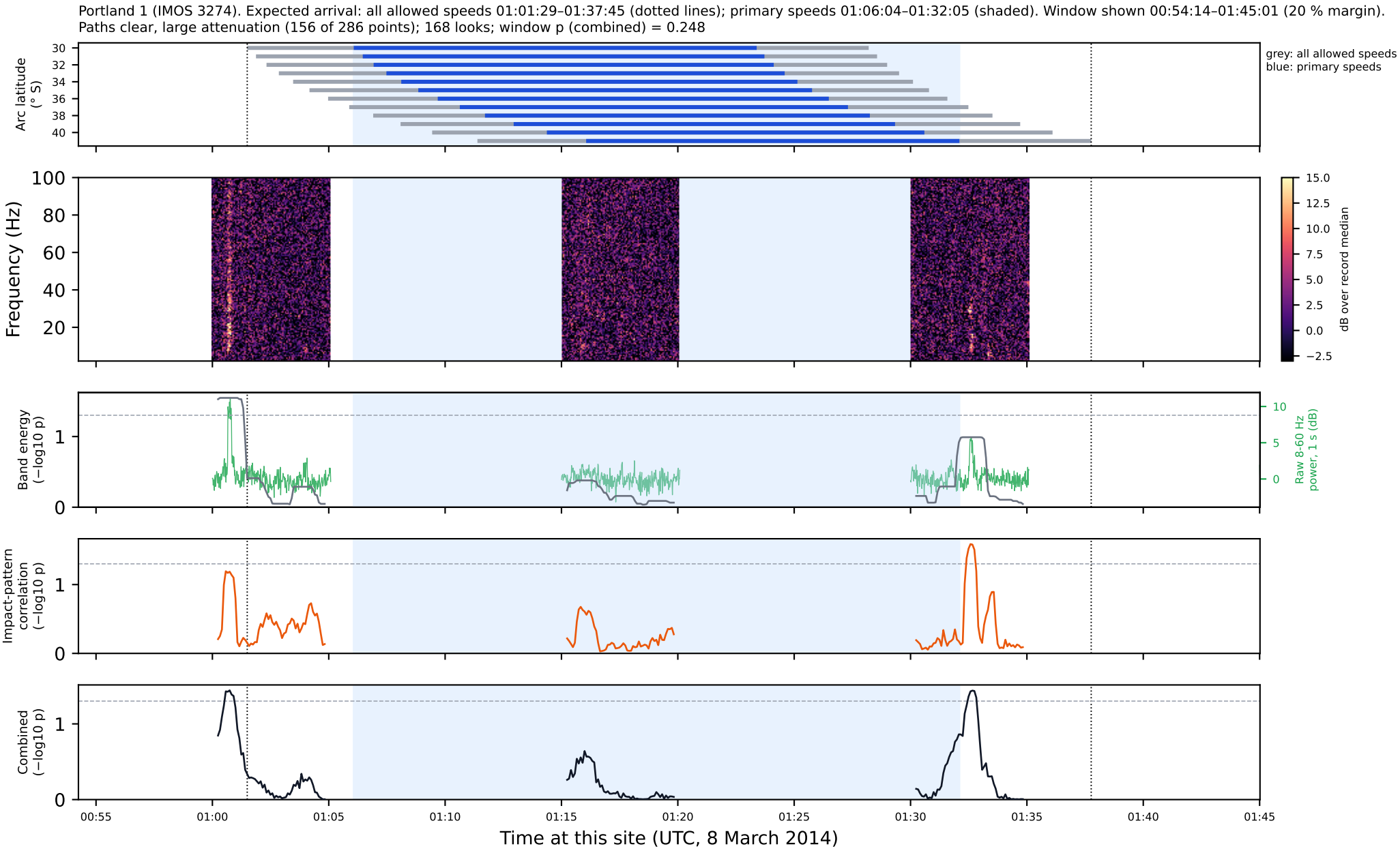
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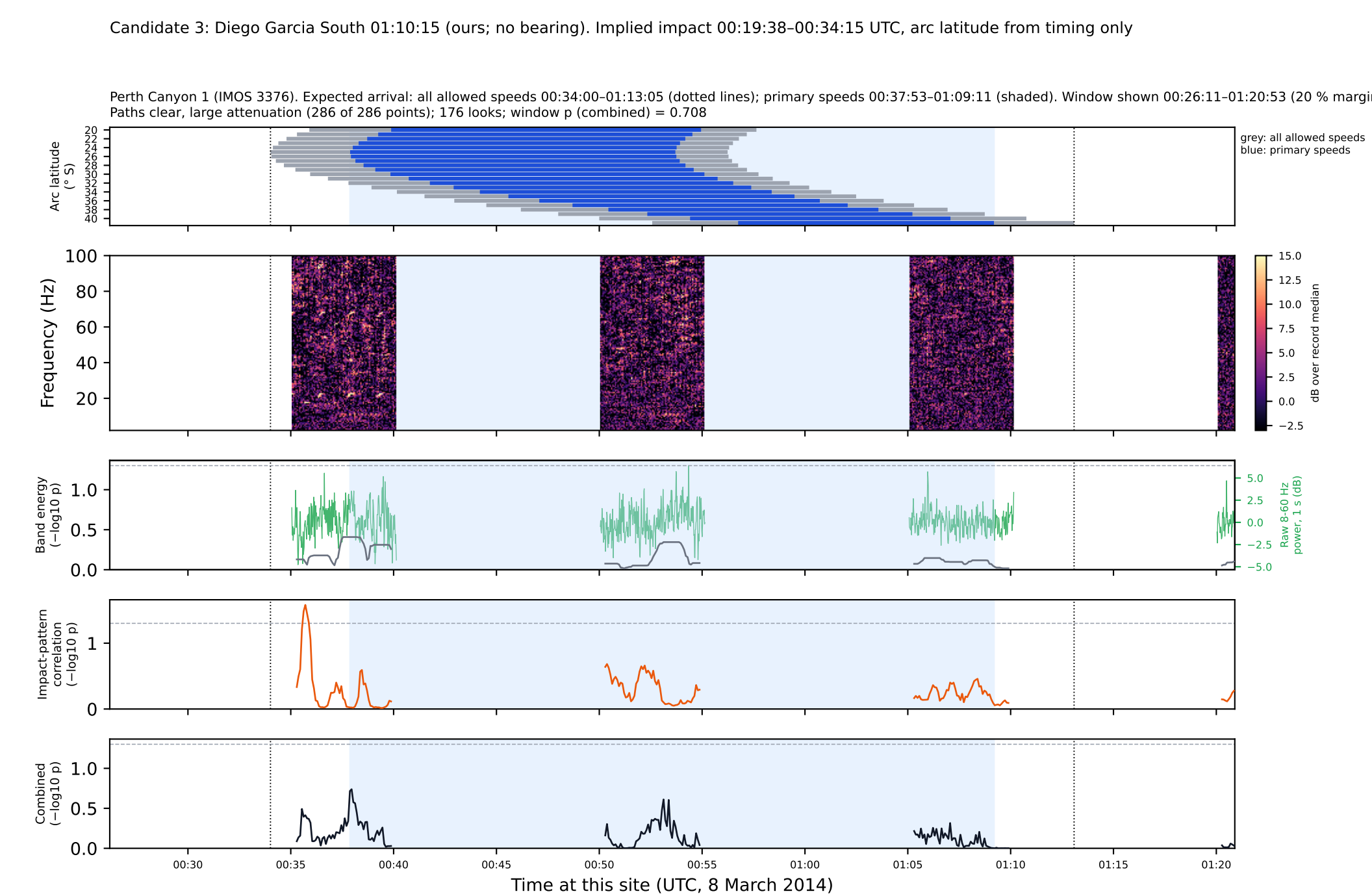


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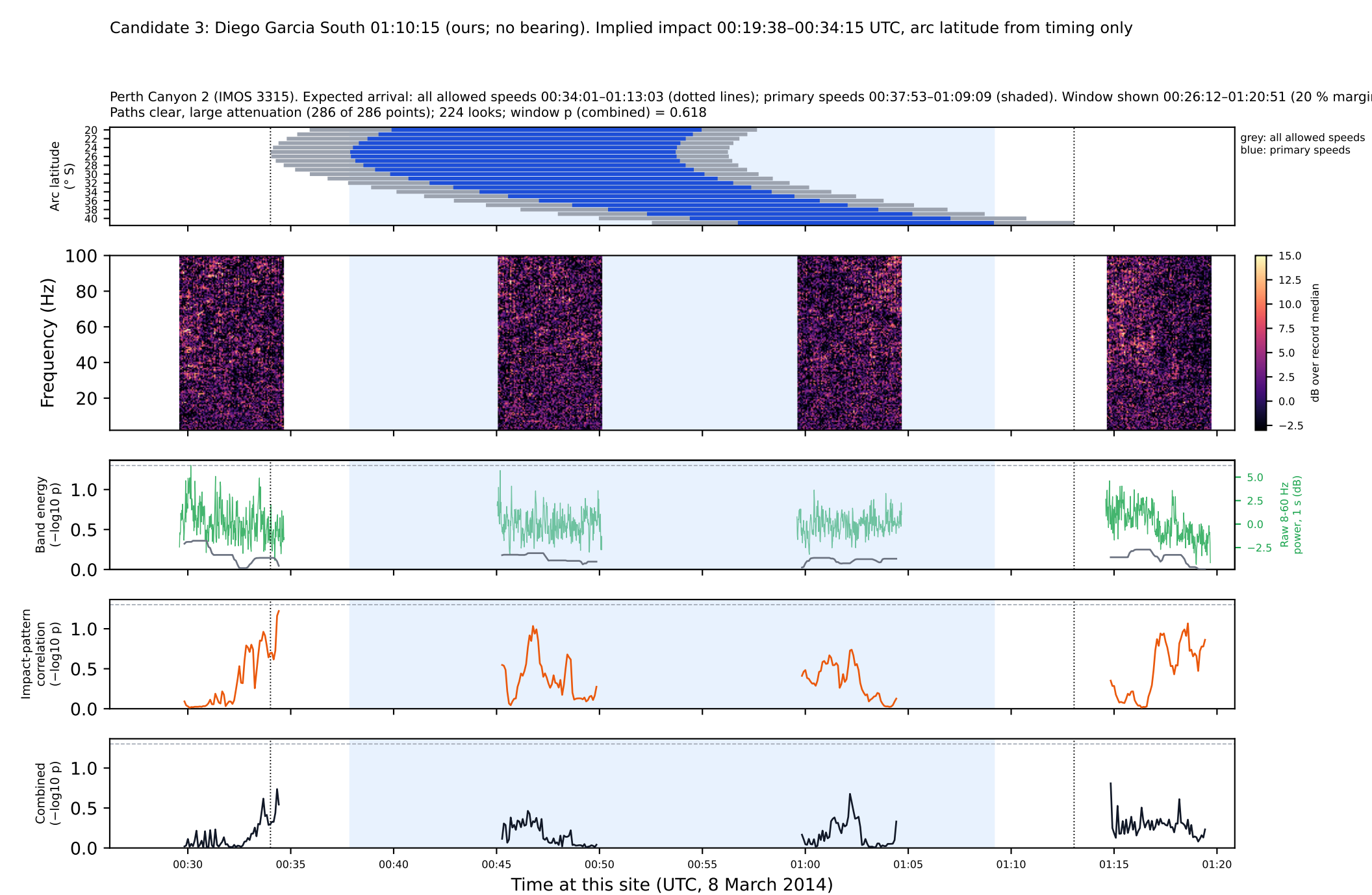
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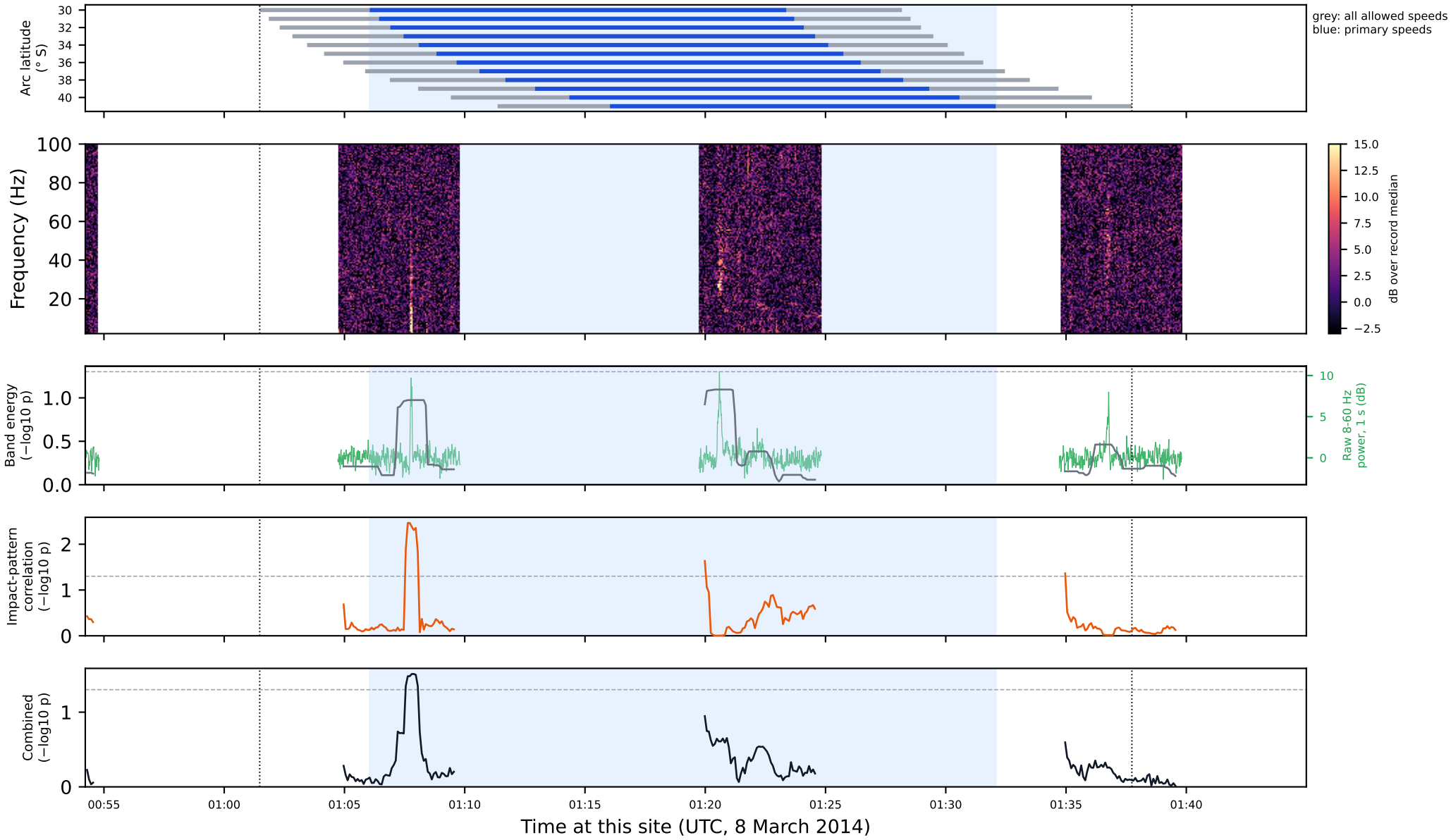
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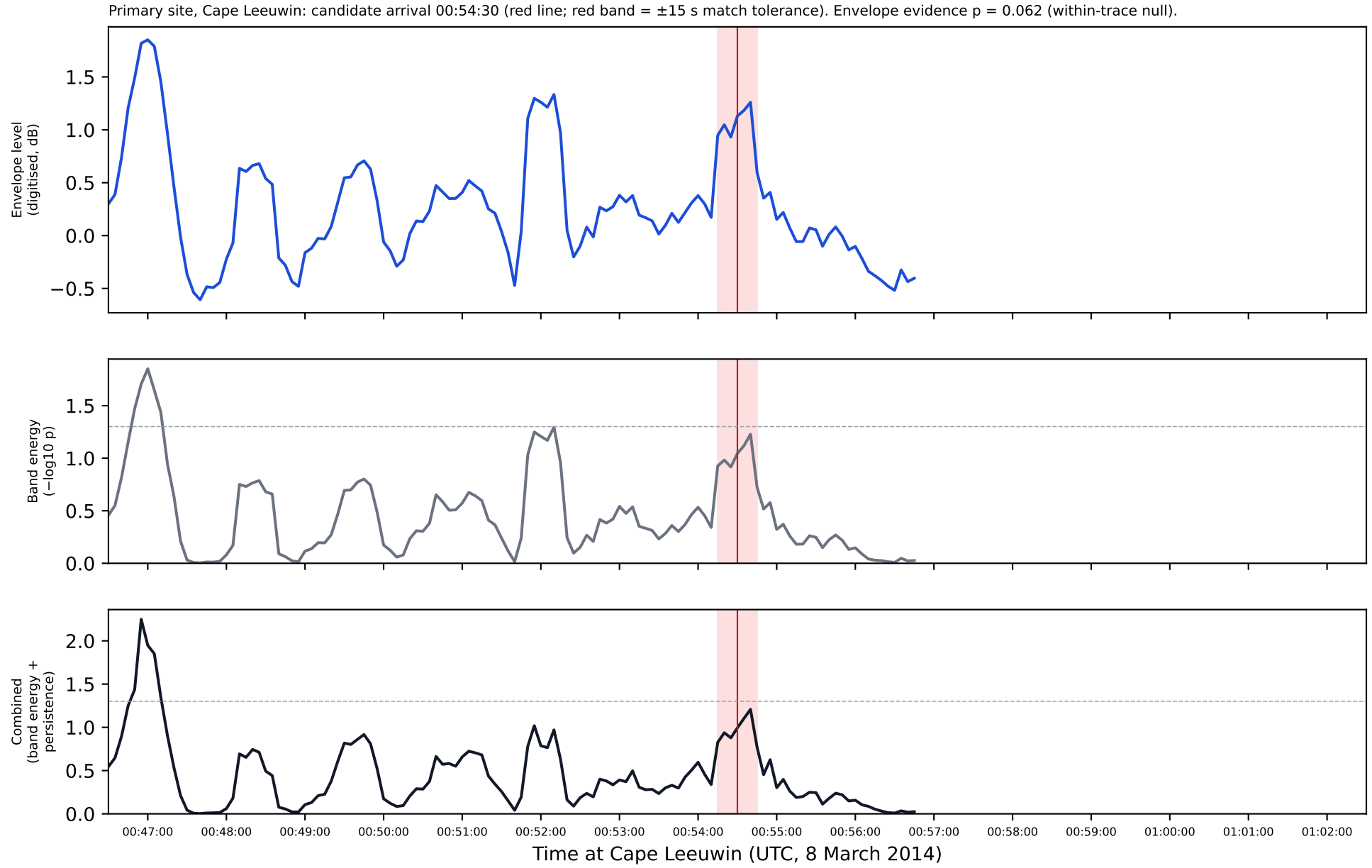
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Portland 2 (IMOS 3275). Expected arrival: all allowed speeds 01:01:28–01:37:44 (dotted lines); primary speeds 01:06:03–01:32:04 (shaded). Window shown 00:54:13–01:44:59 (20 % margin).
Paths clear, large attenuation (156 of 286 points); 172 looks; window p (combined) = 0.176



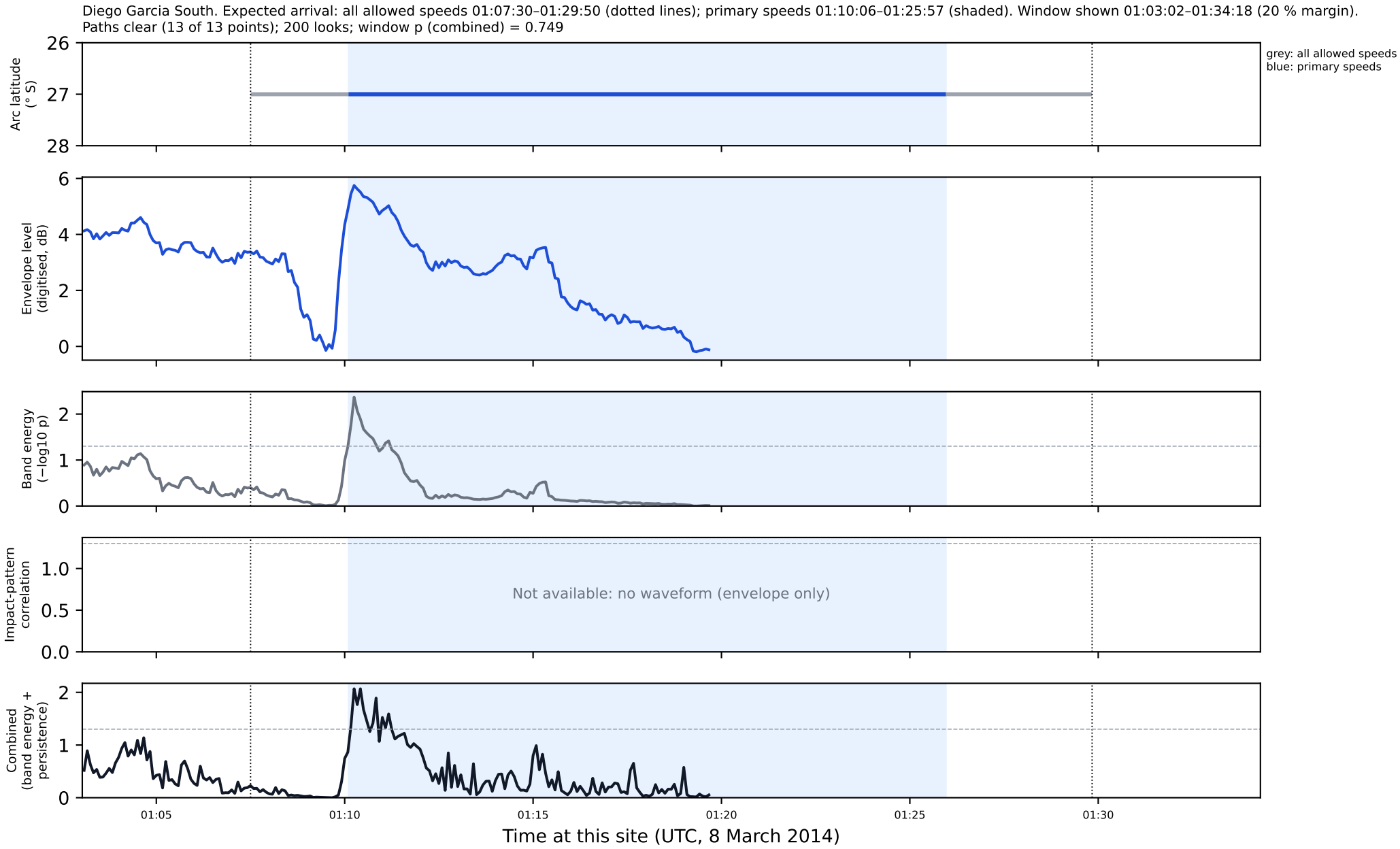
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Candidate 4: Cape Leeuwin 00:54:30 (Kadri and ours; bearing 299.4° corrected, 306.2° published). Implied impact 00:31:26–00:40:42 UTC, arc 27° S



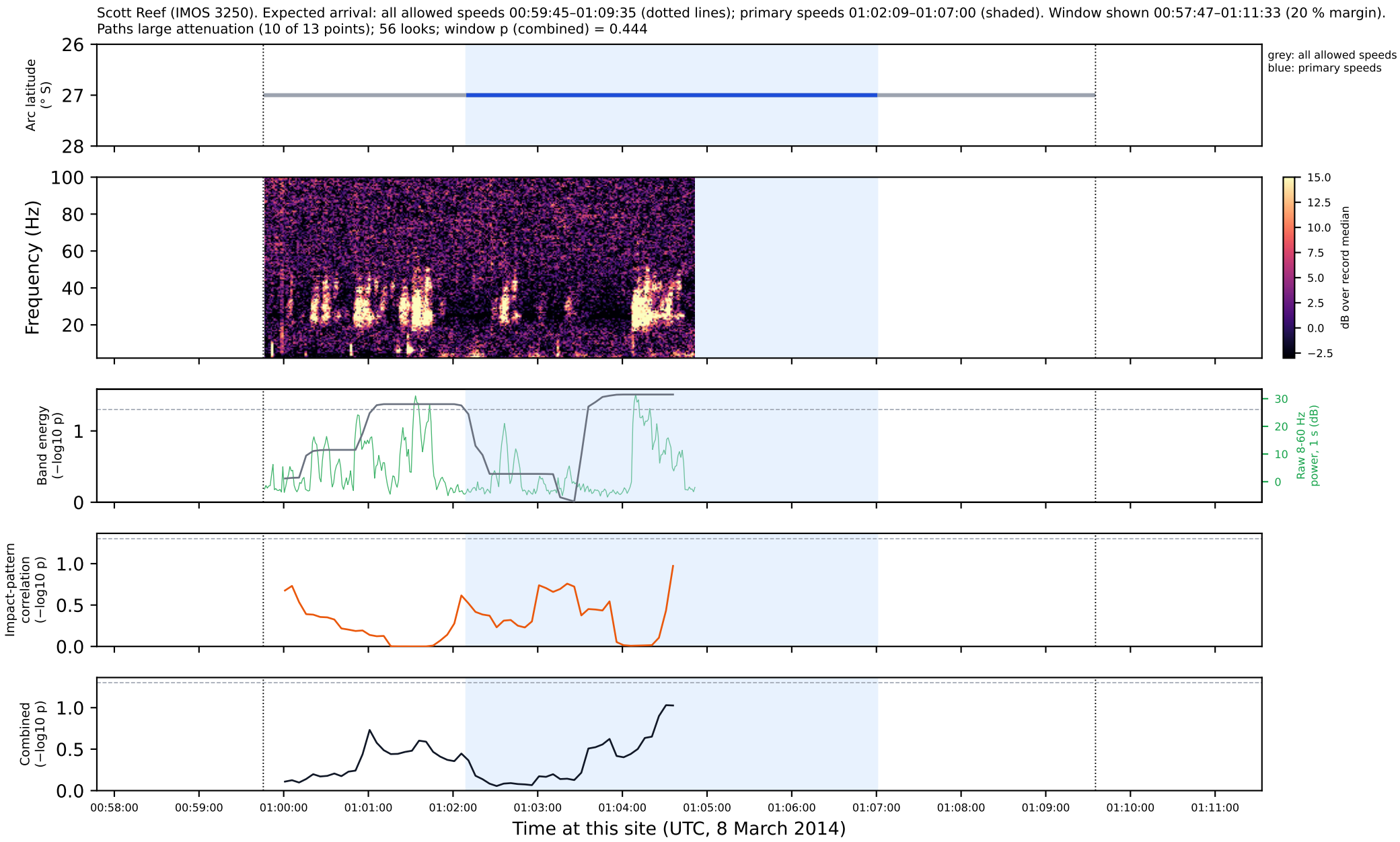
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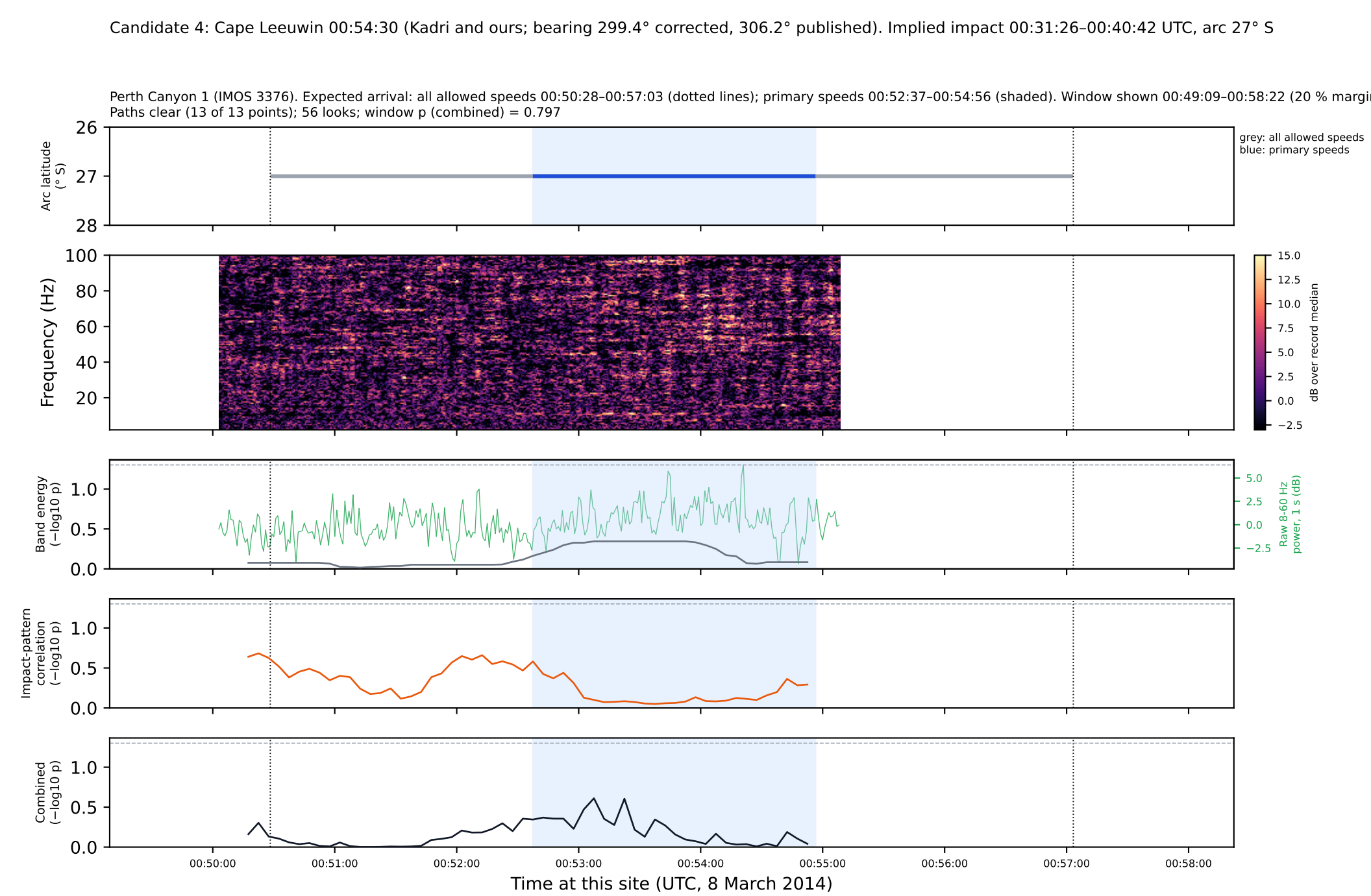


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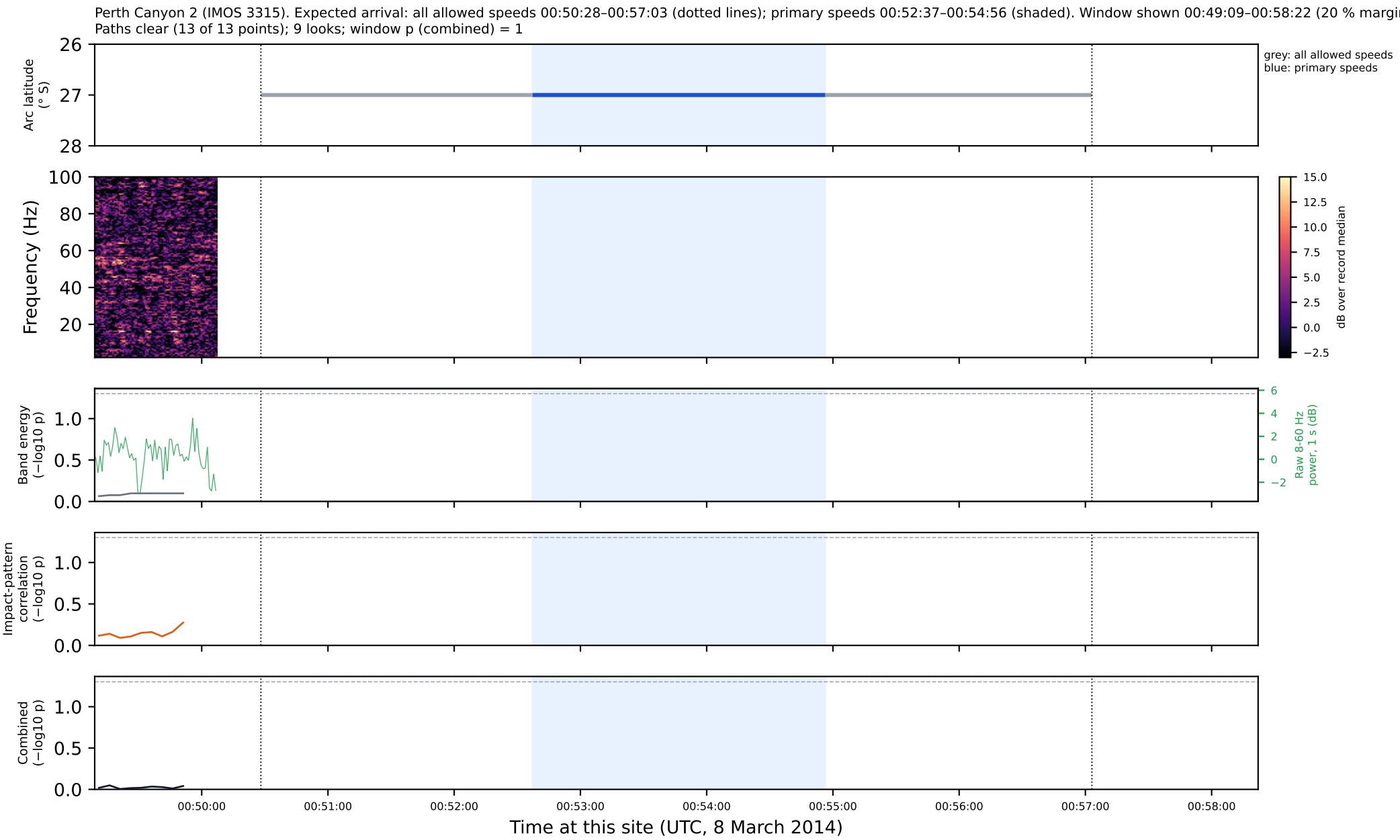


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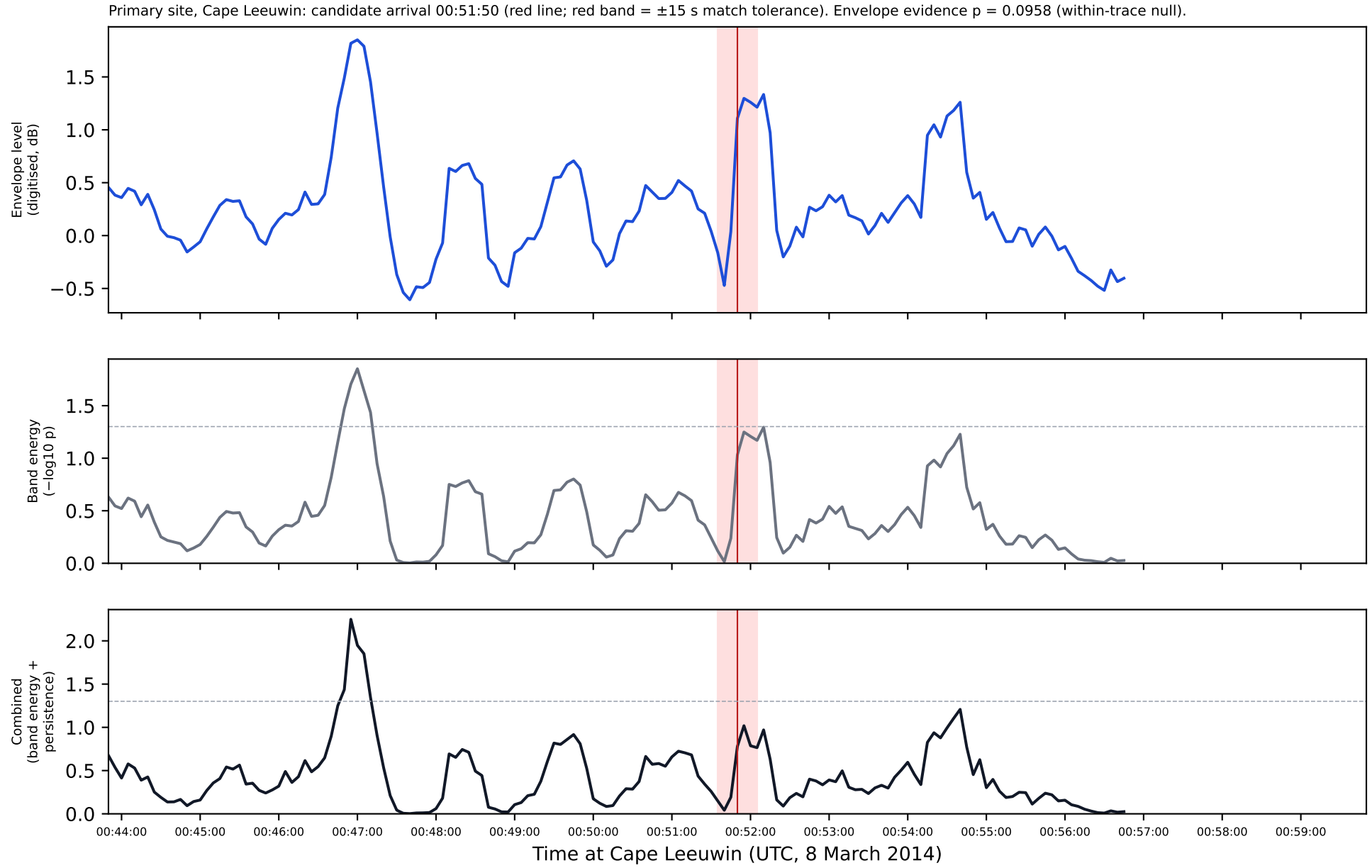
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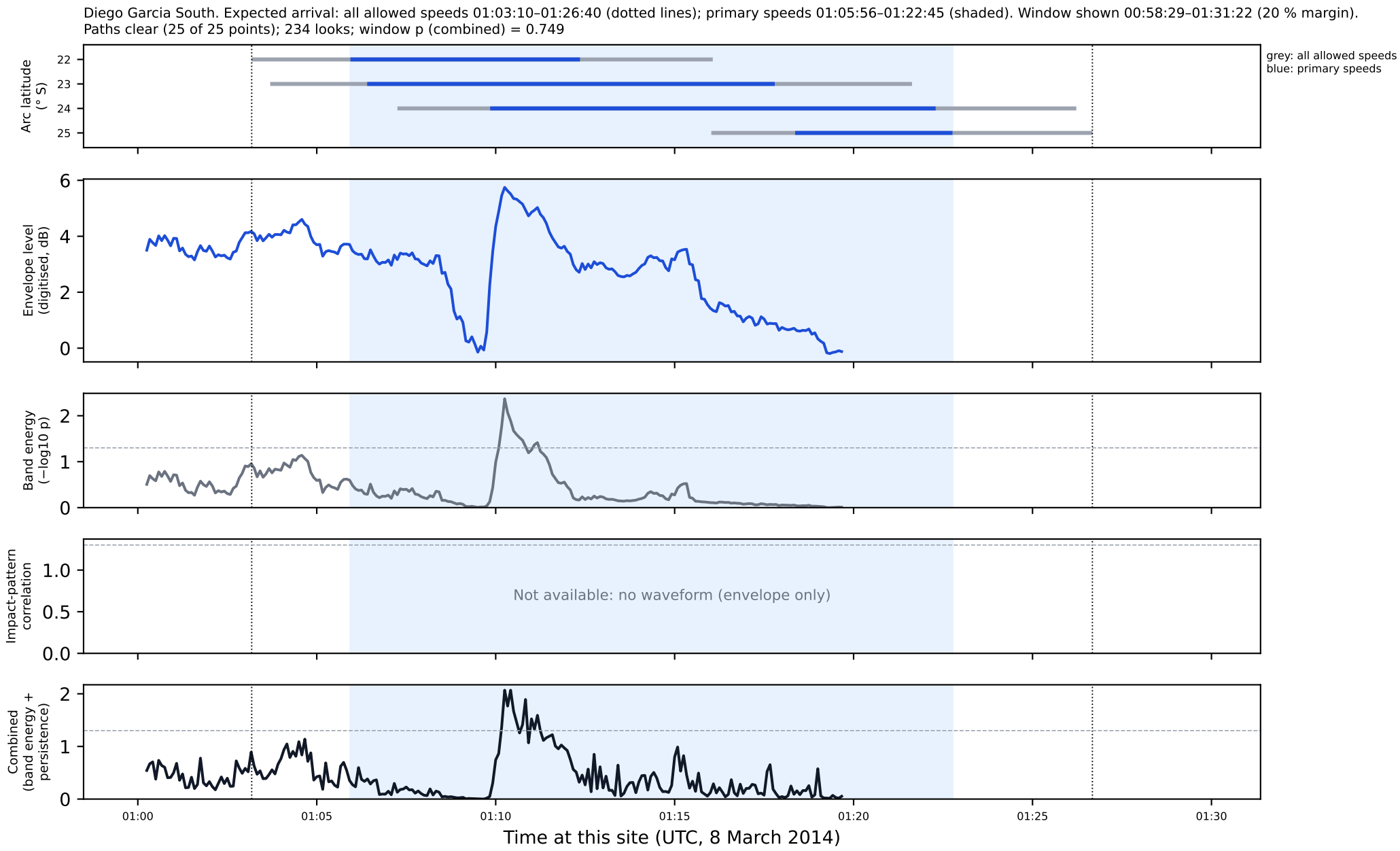
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Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:26:44–00:37:25 UTC, arc 23; 24° S



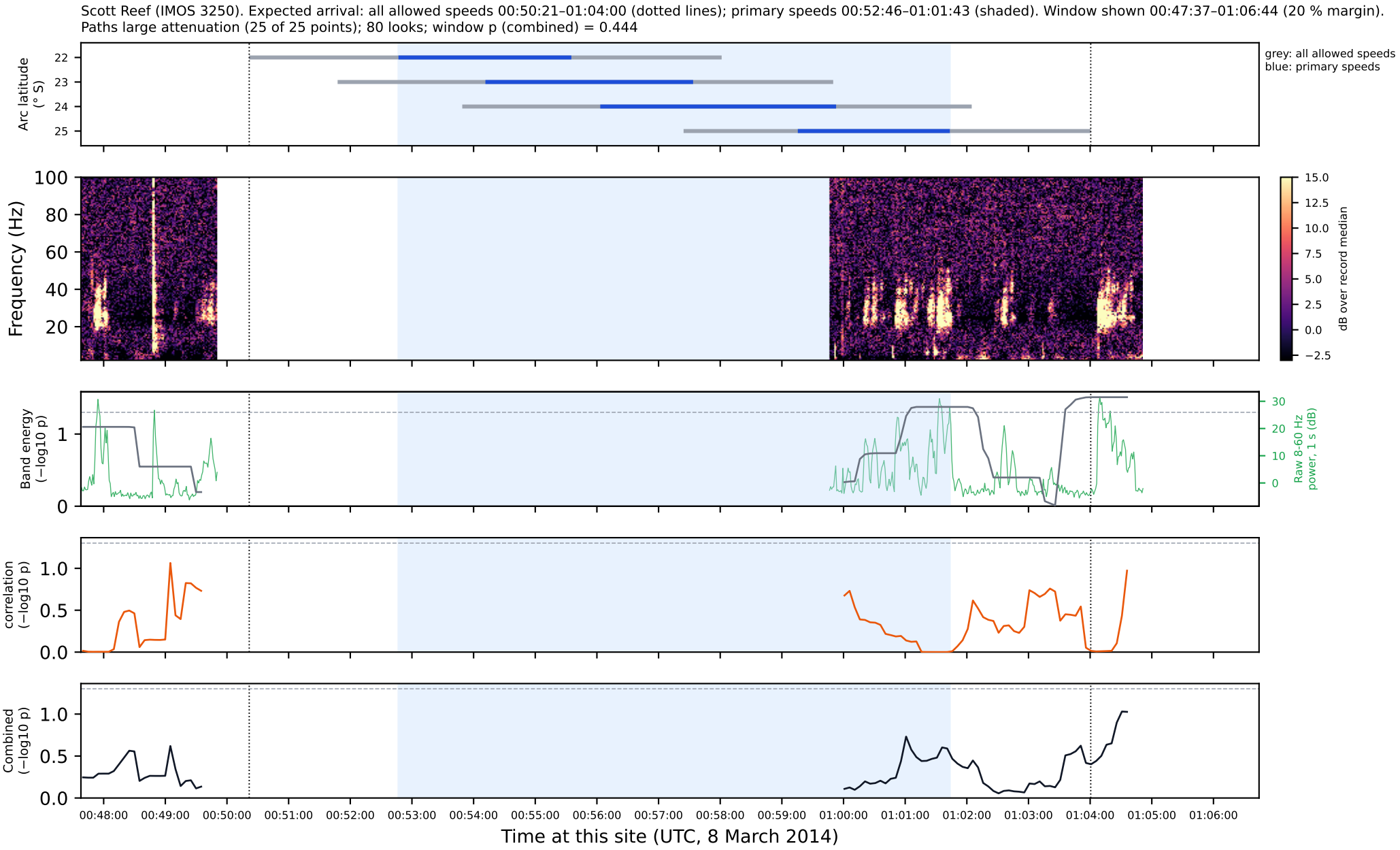
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Candidate 5: Cape Leeuwin 00:51:50 (Kadri and ours; bearing 315.5° corrected, 320.4° published). Implied impact 00:26:44–00:37:25 UTC, arc 23; 24° S



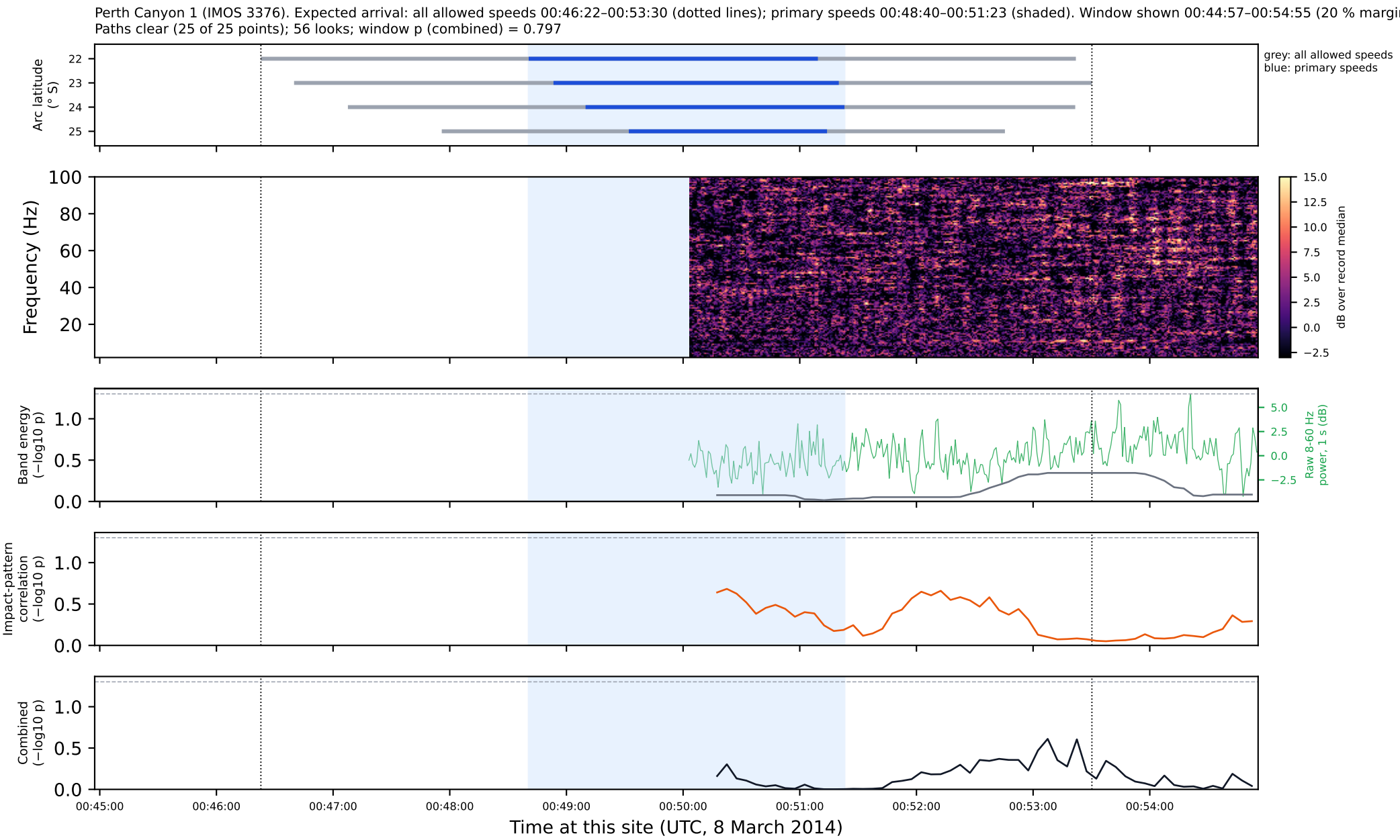
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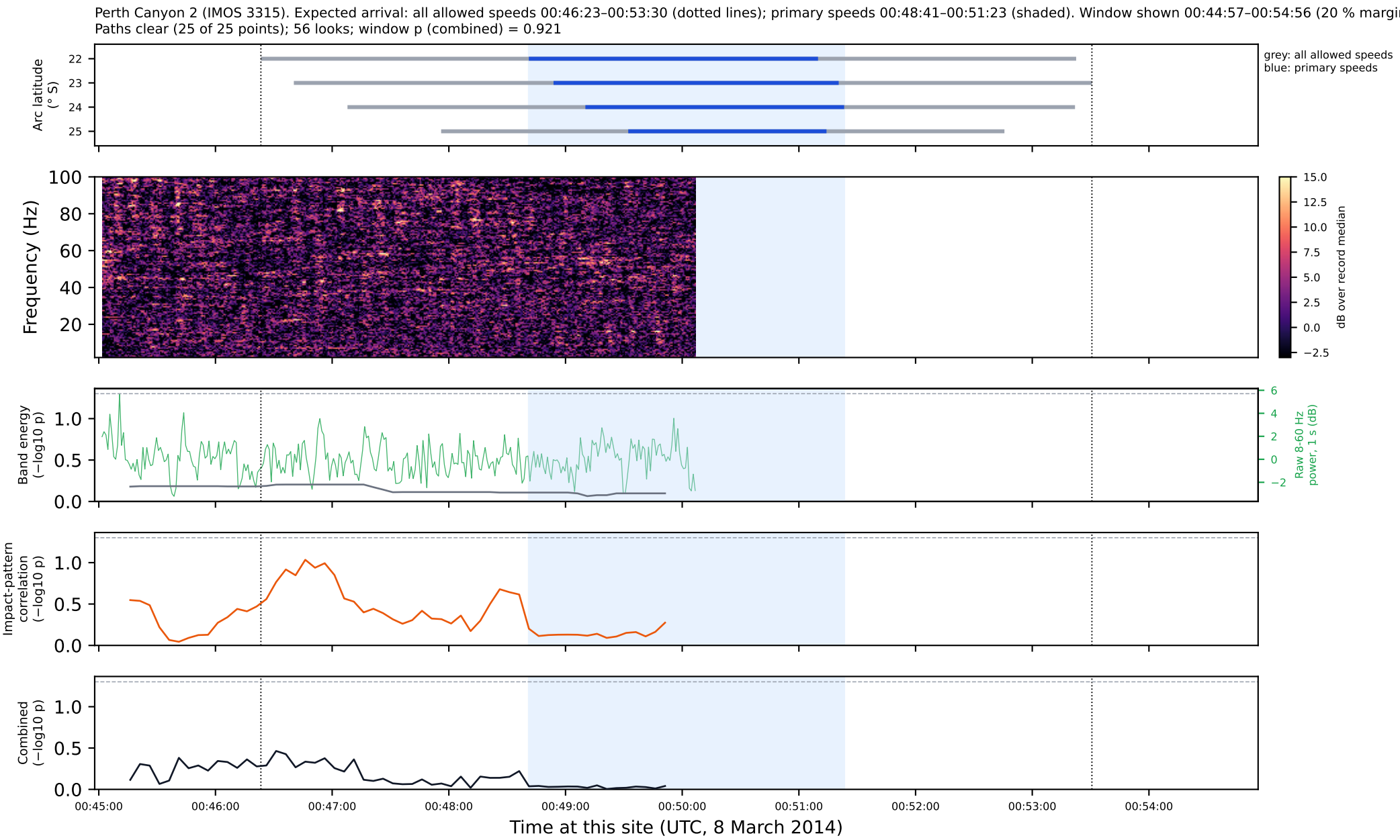
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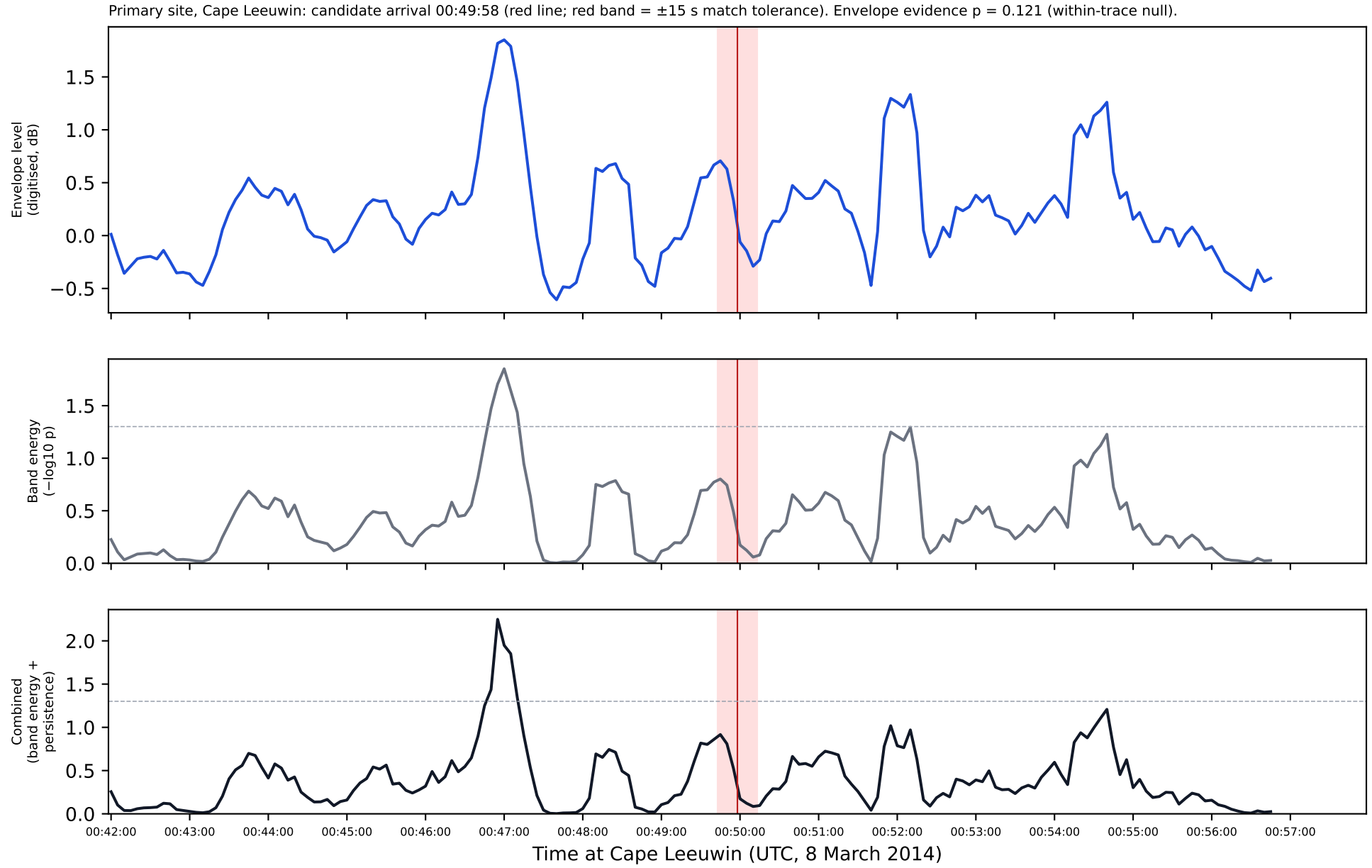
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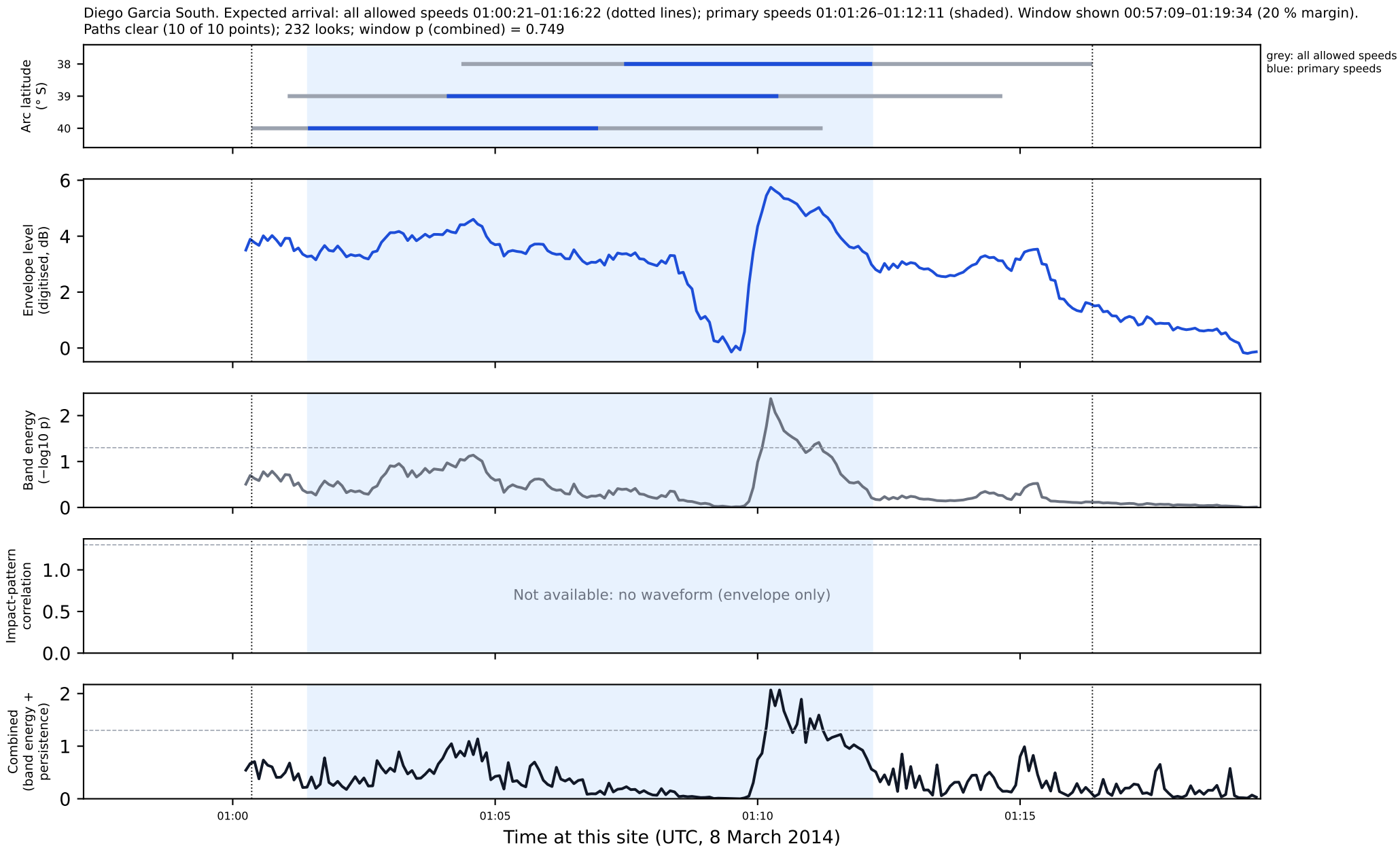
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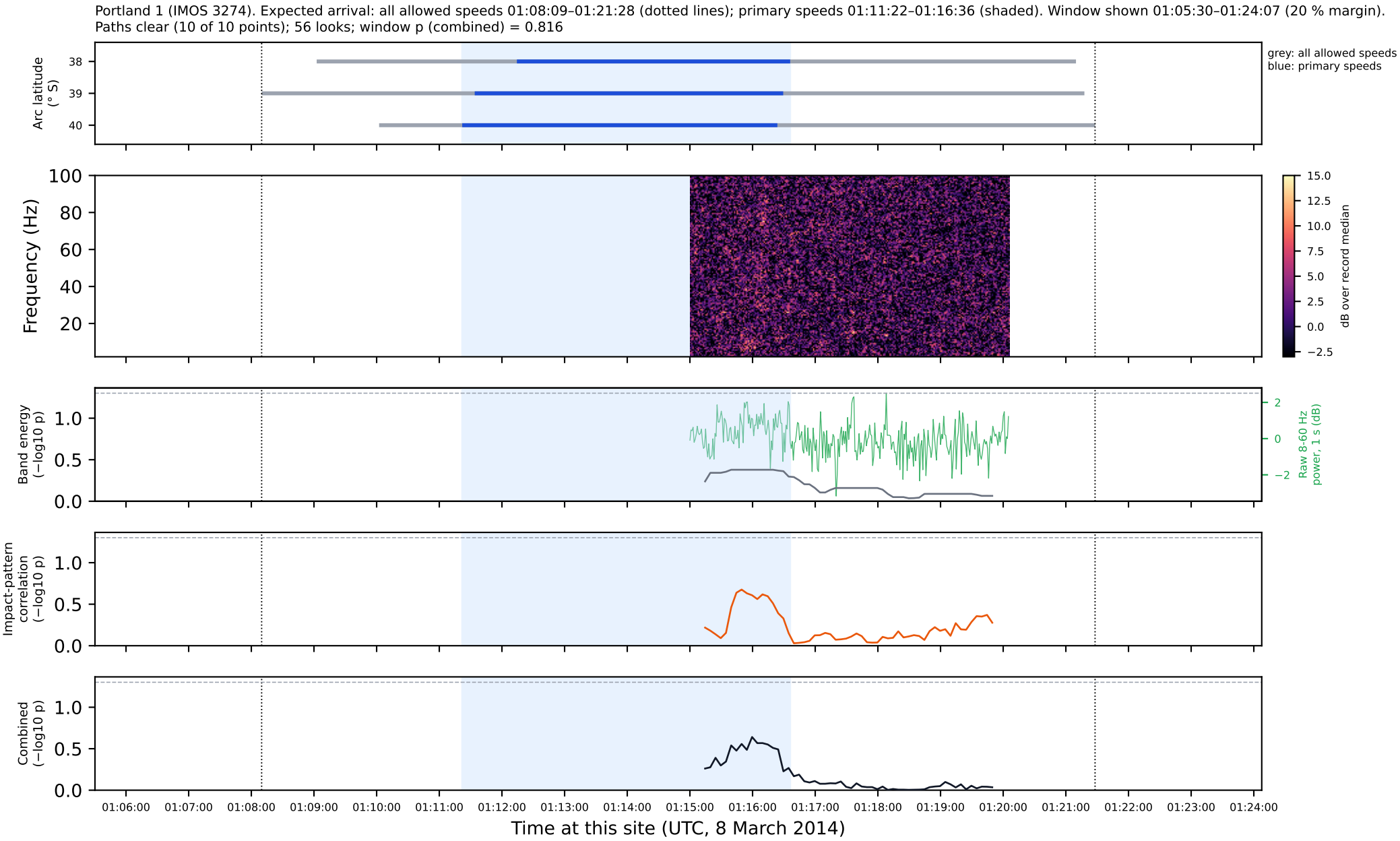
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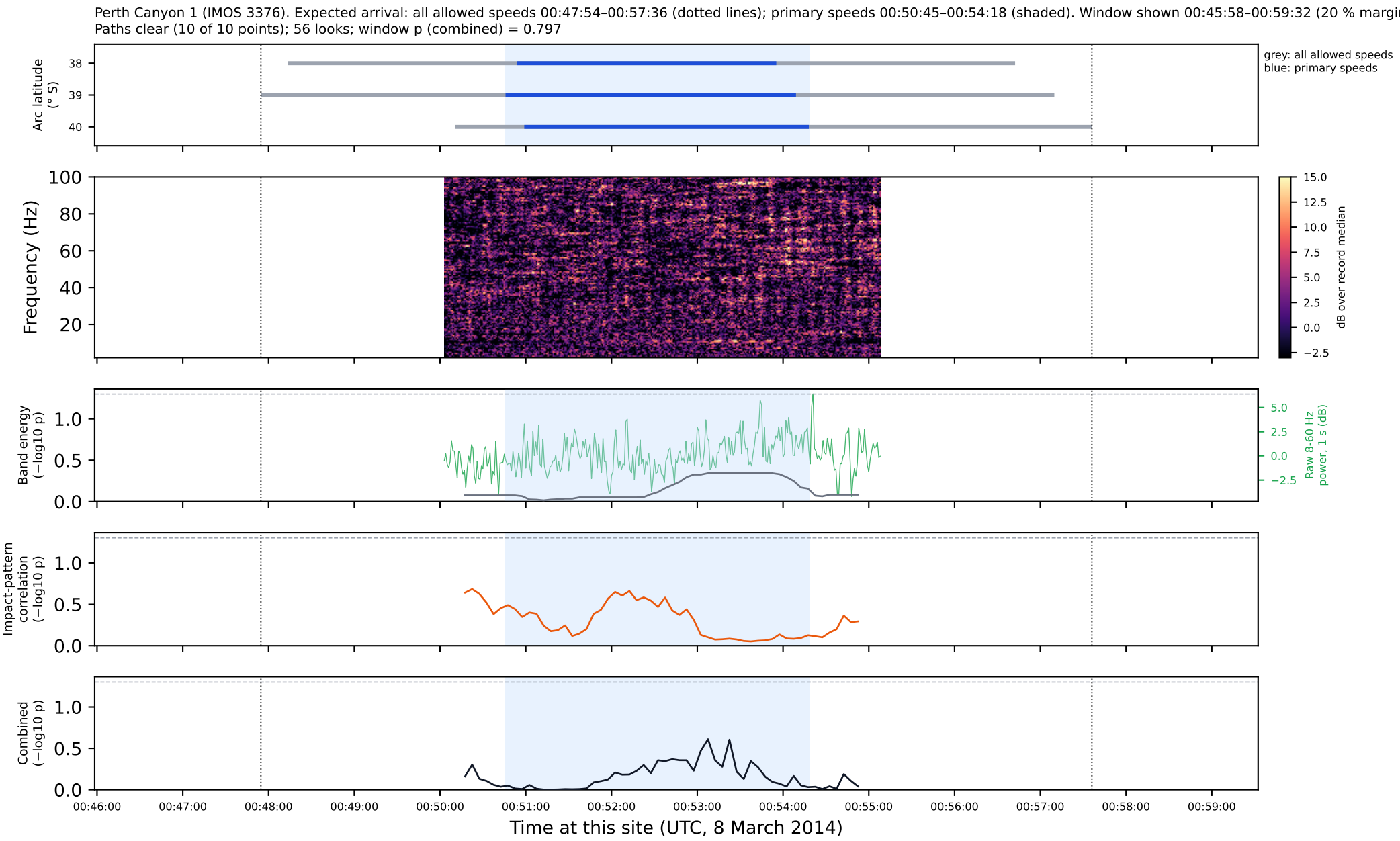
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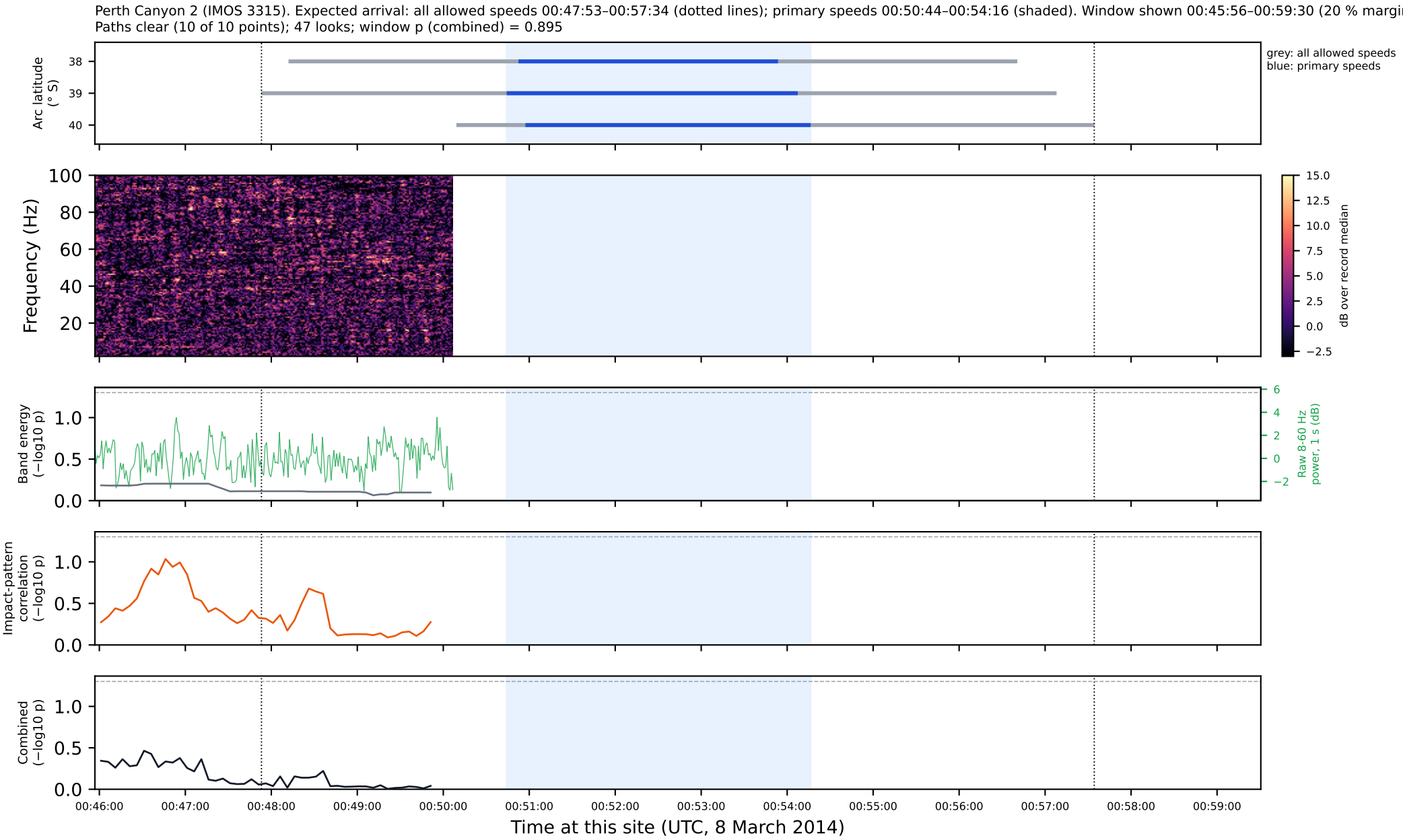
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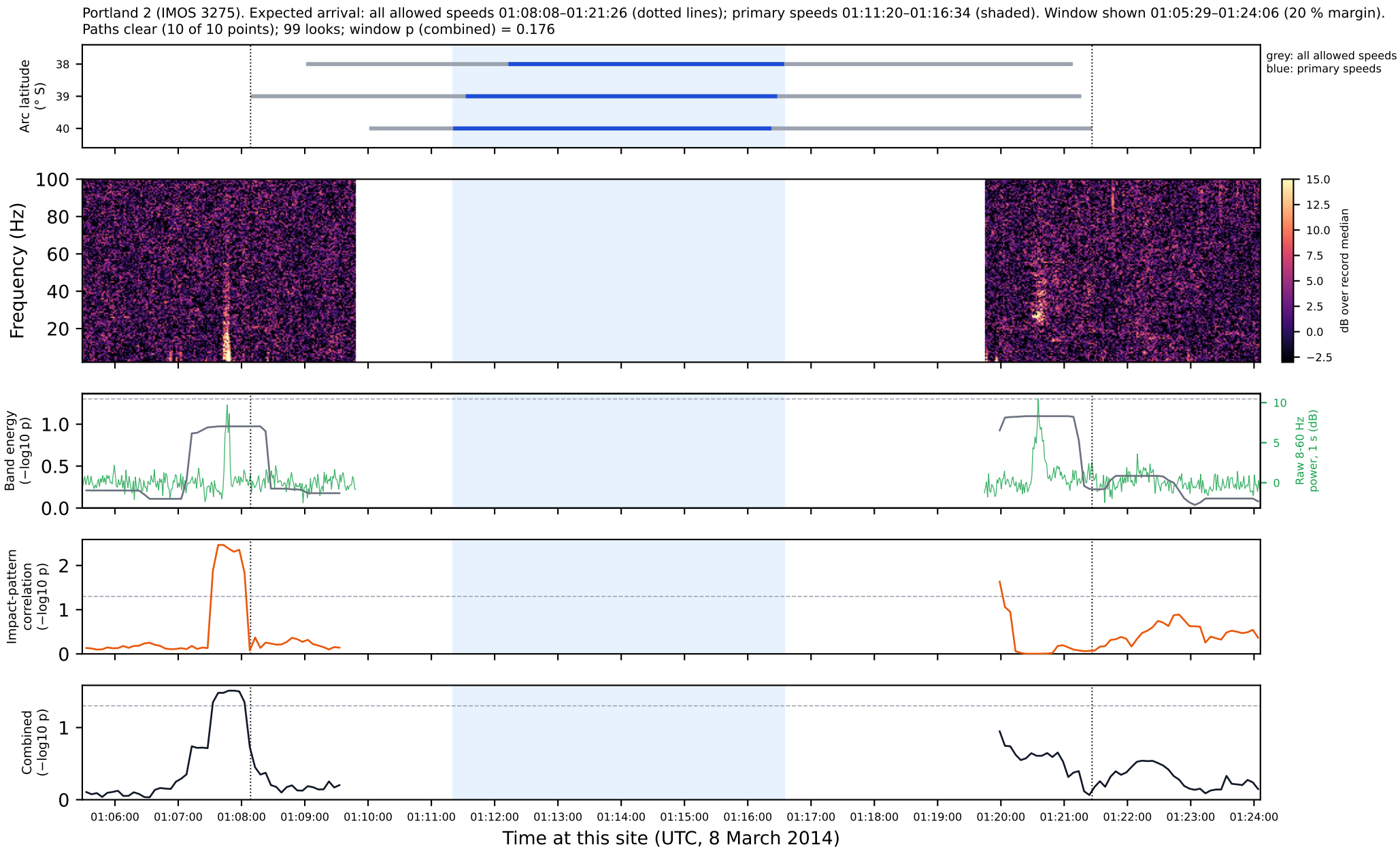
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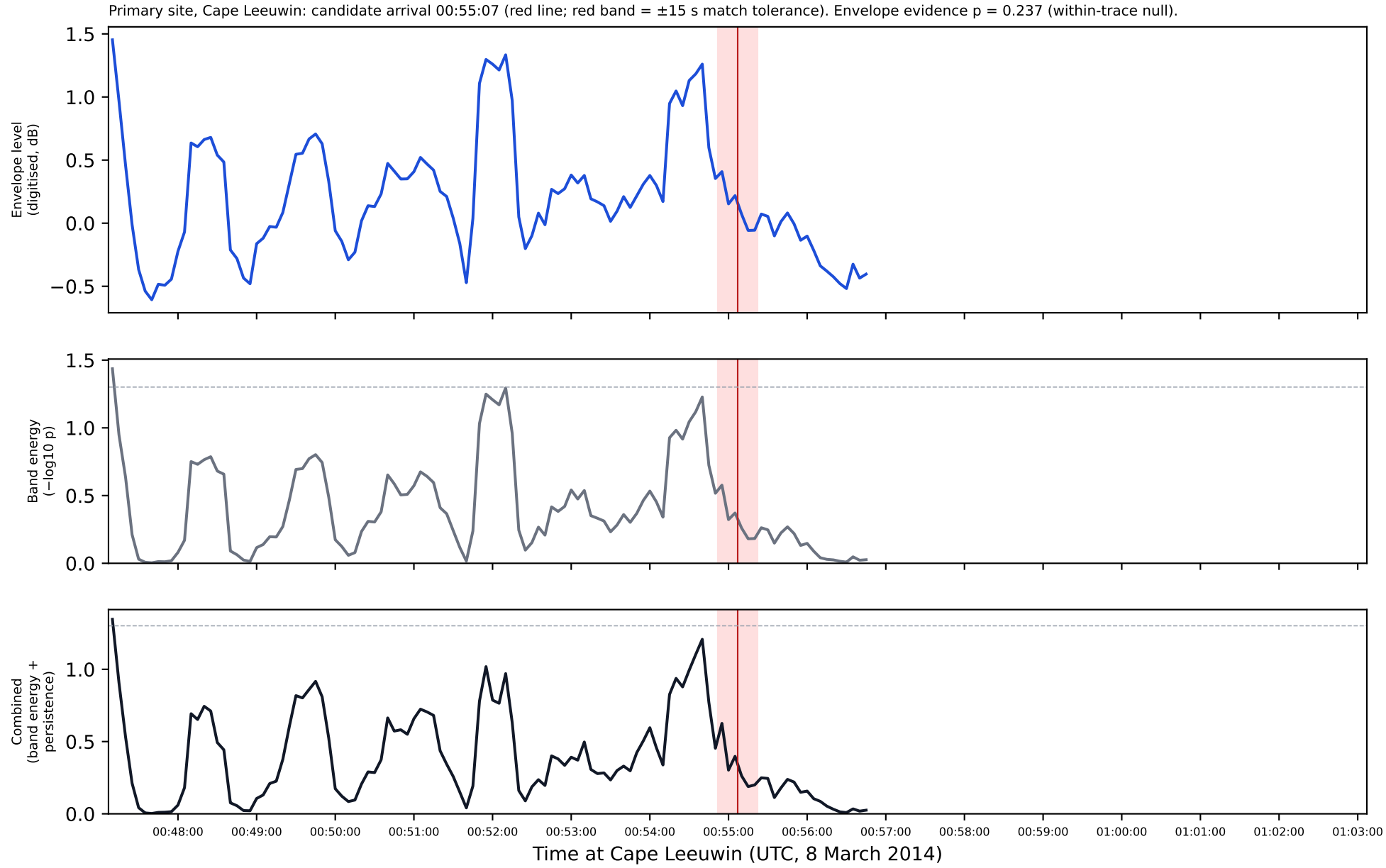
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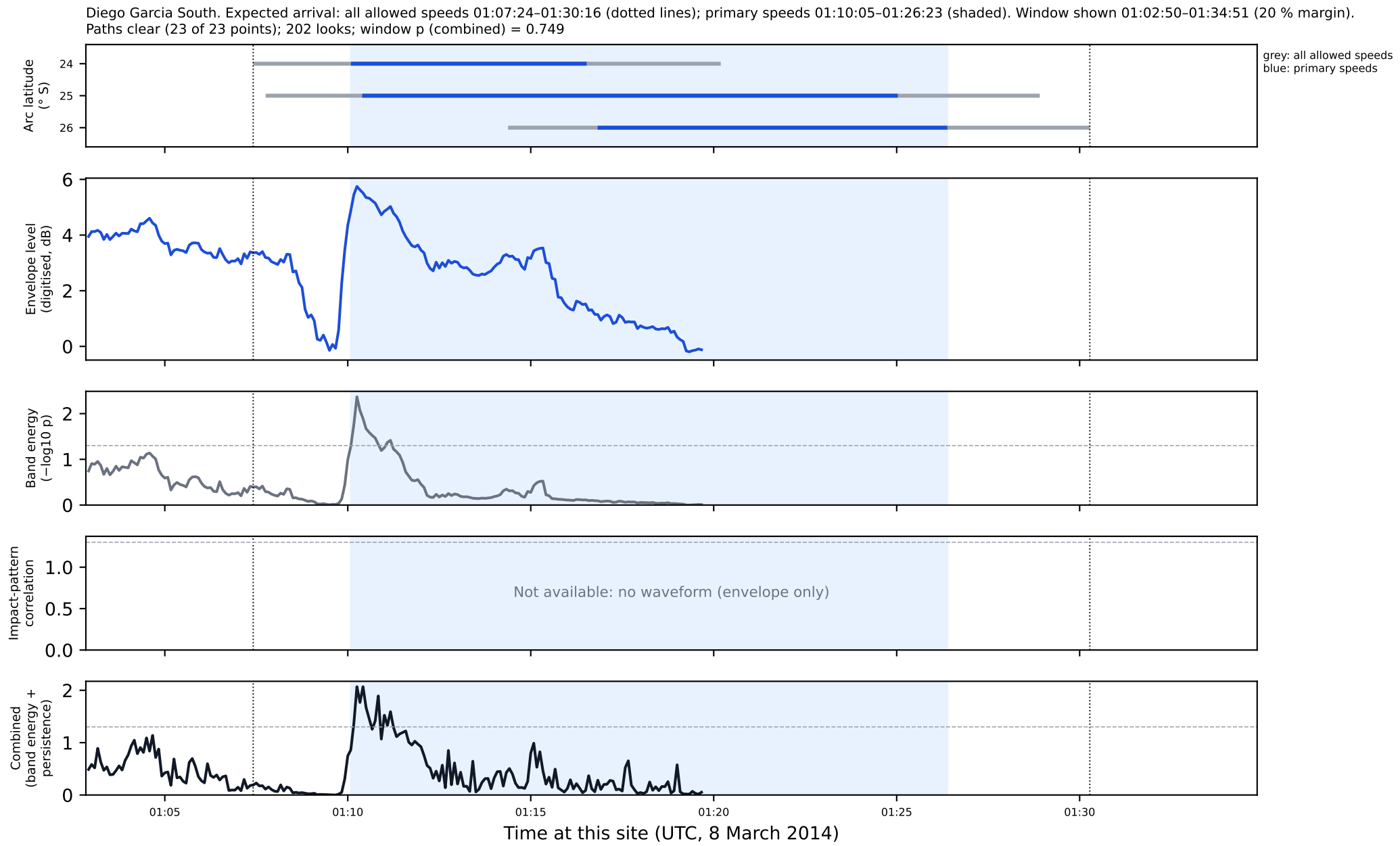
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Candidate 7: Cape Leeuwin 00:55:07 (Kadri; bearing 308.5° corrected, 314.1° published). Implied impact 00:31:10–00:41:05 UTC, arc 25; 26° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

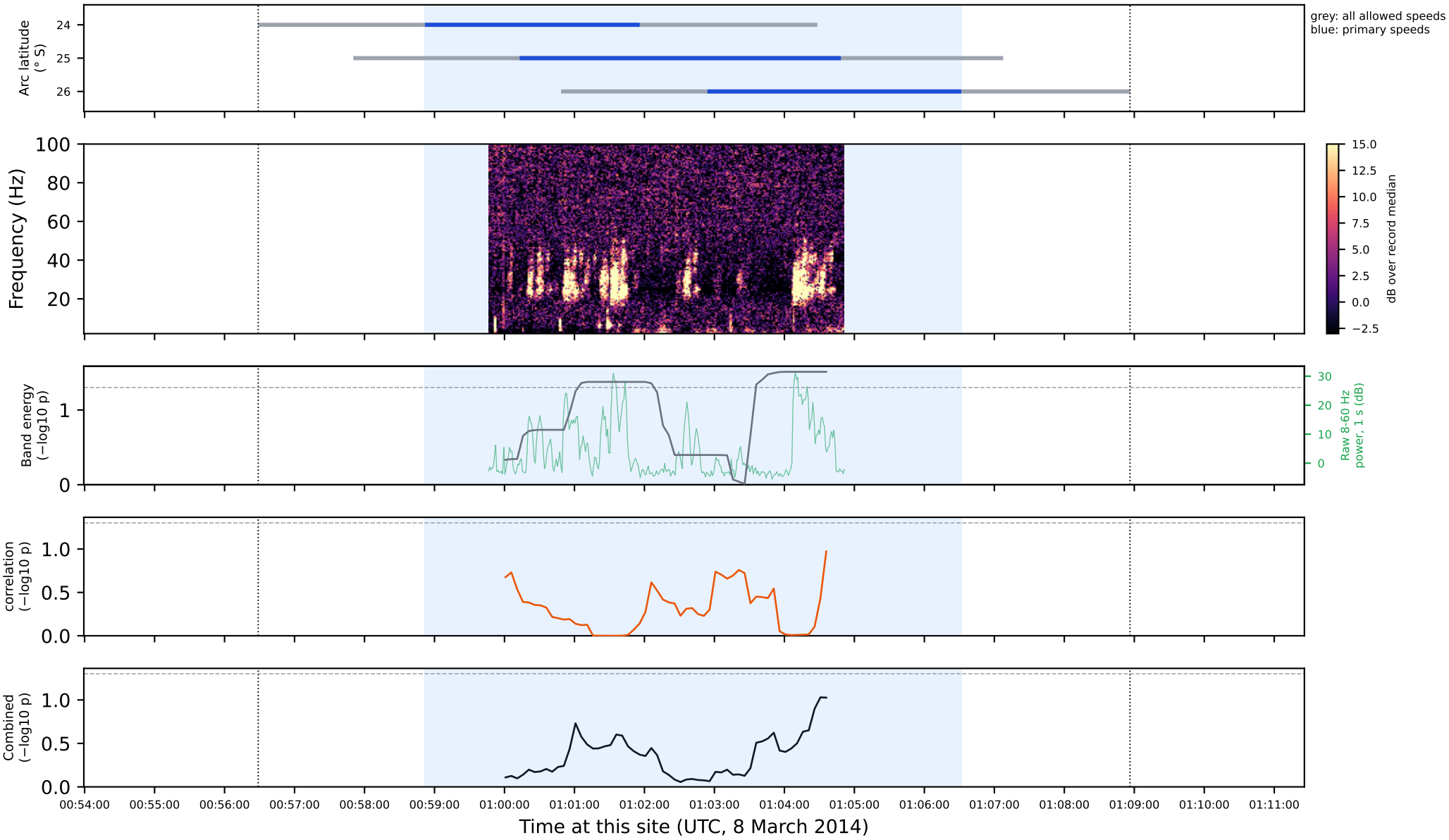
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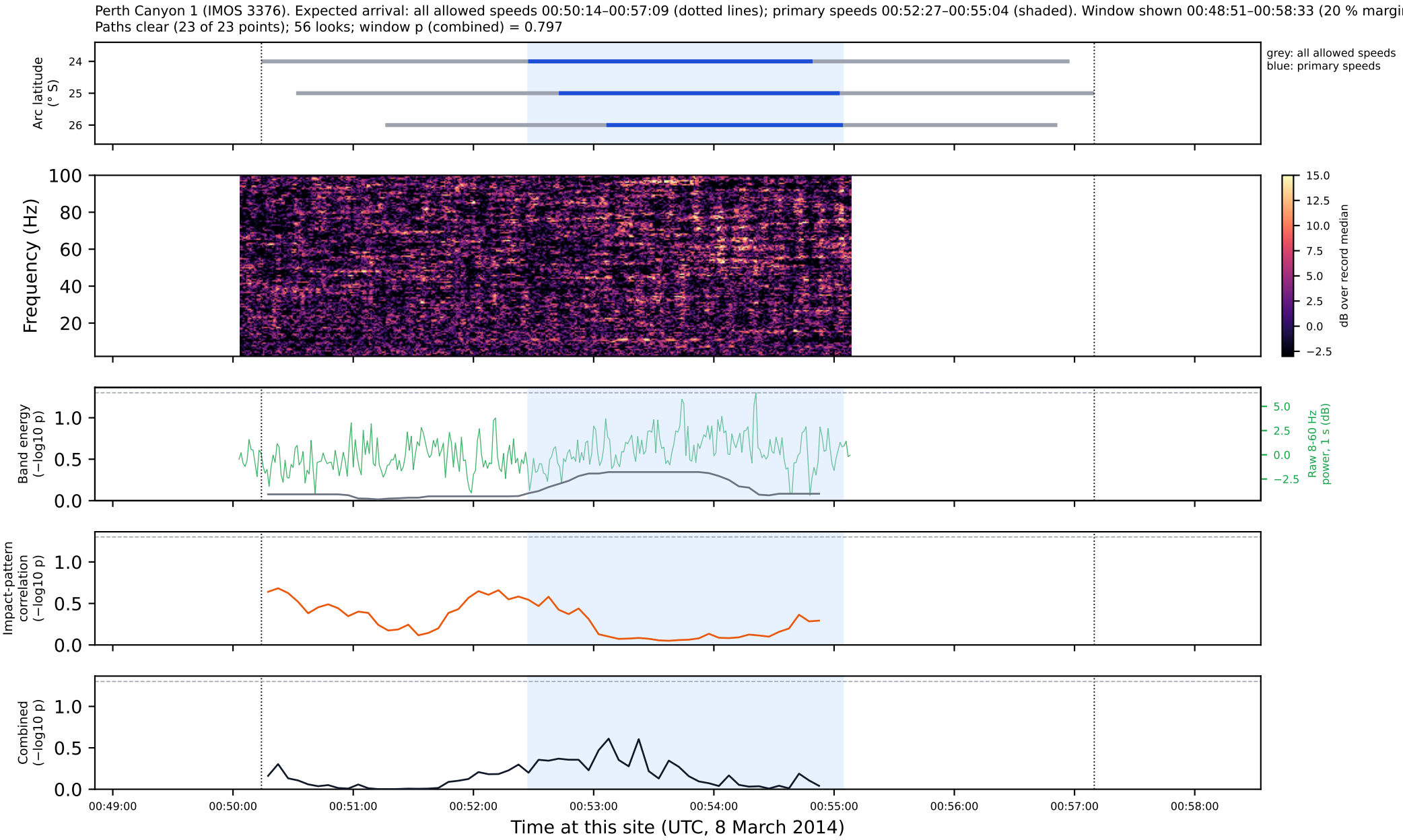
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Scott Reef (IMOS 3250). Expected arrival: all allowed speeds 00:56:28–01:08:56 (dotted lines); primary speeds 00:58:51–01:06:31 (shaded). Window shown 00:53:59–01:11:25 (20 % margin).
Paths large attenuation (22 of 23 points); 56 looks; window p (combined) = 0.444



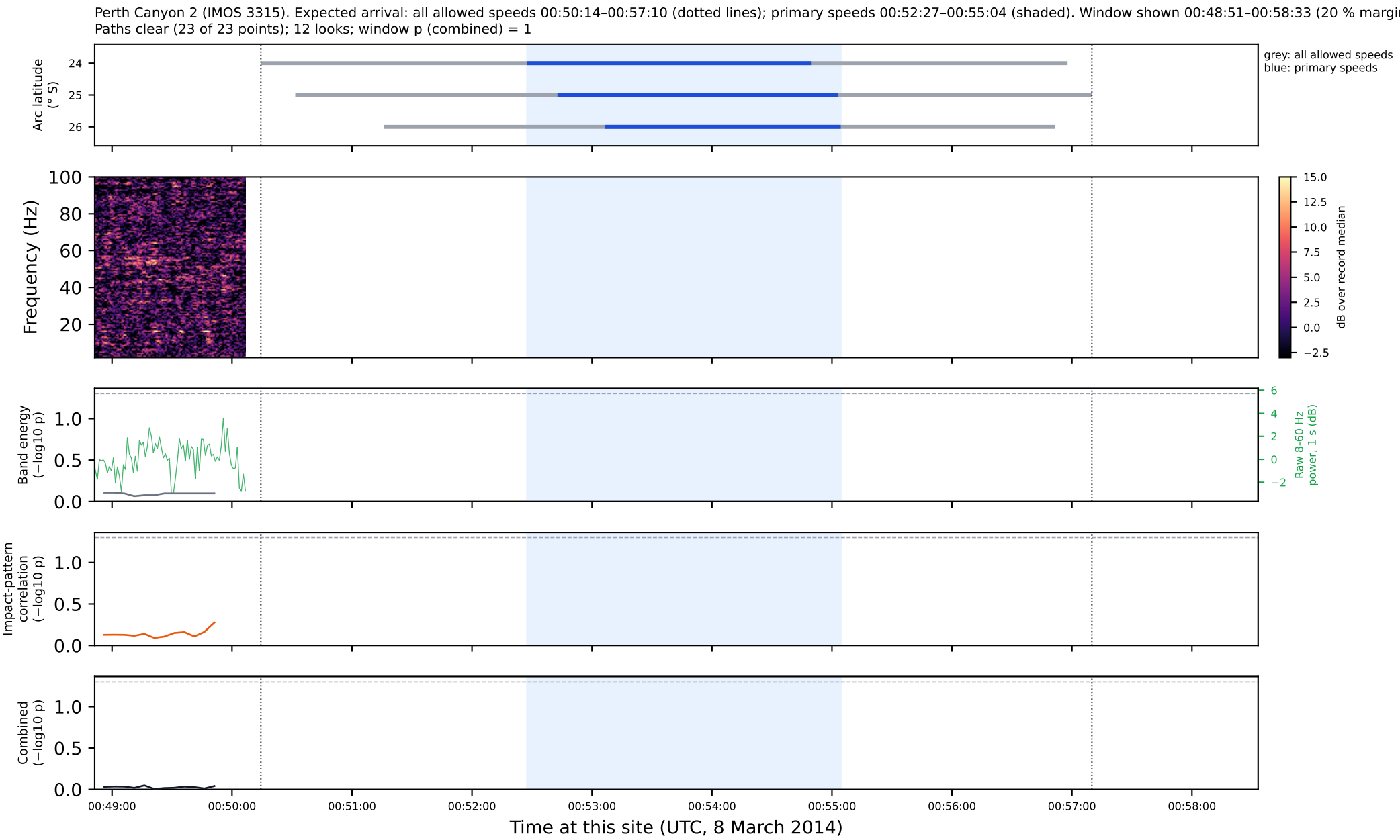
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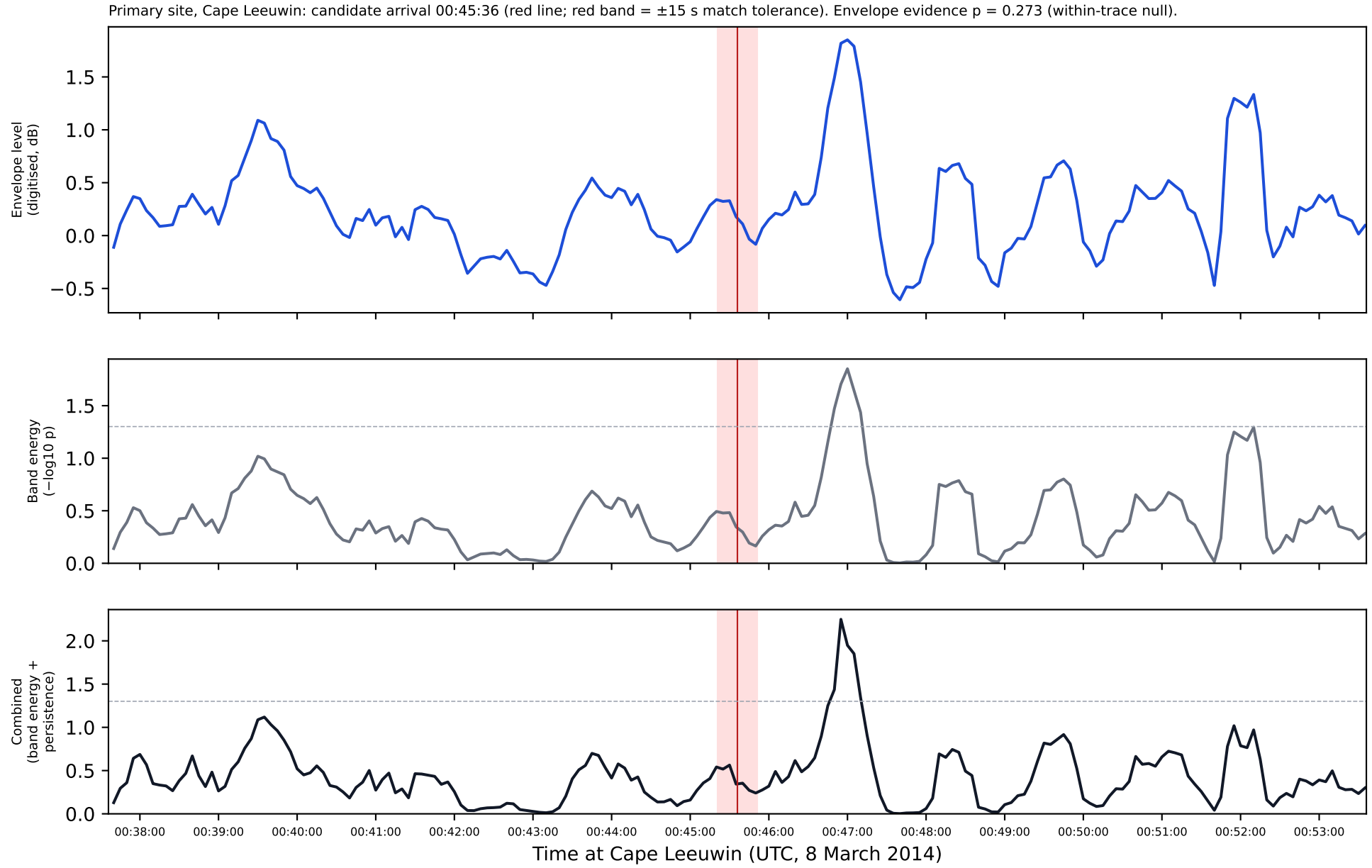
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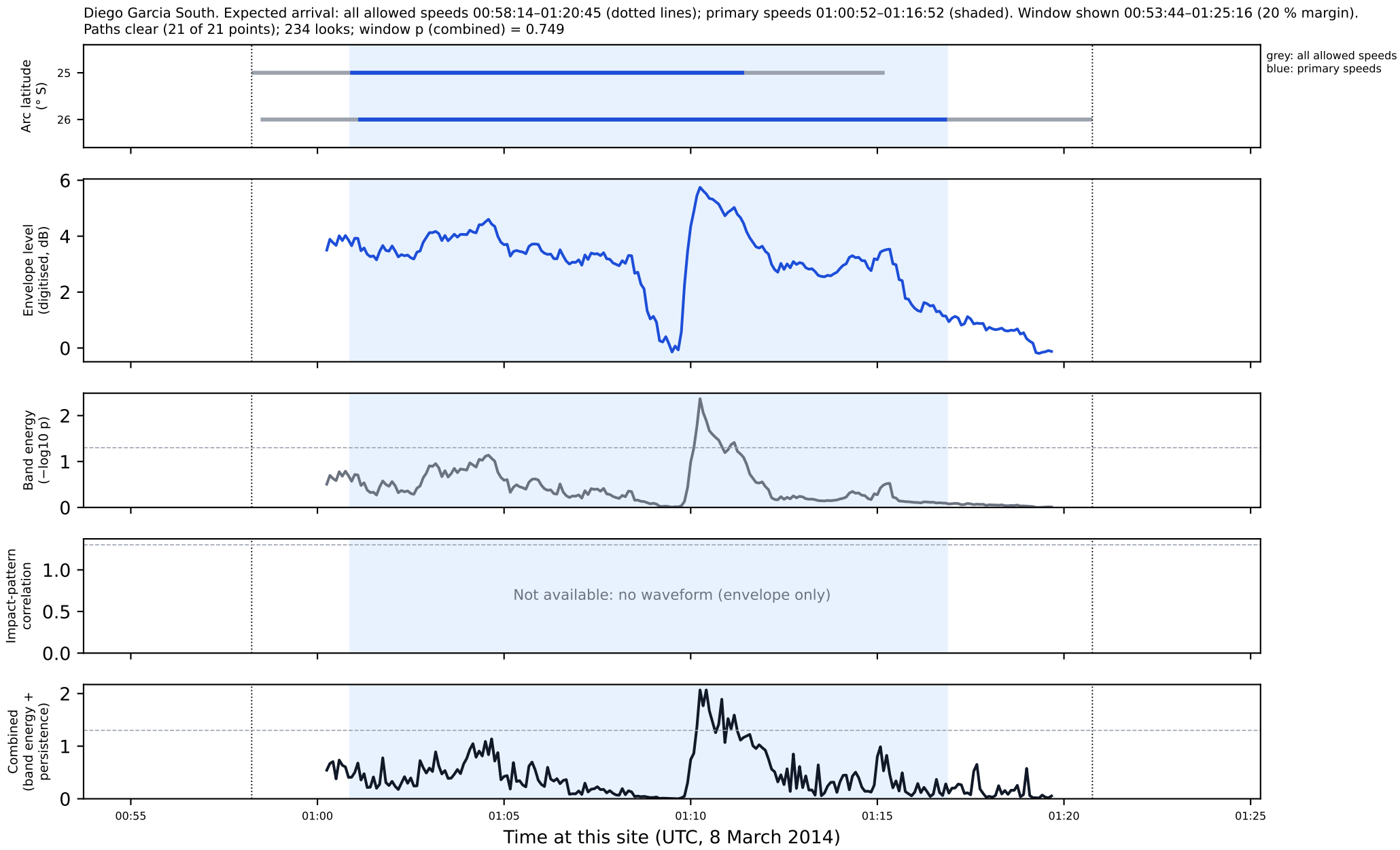
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Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:22:04–00:31:34 UTC, arc 25; 26° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

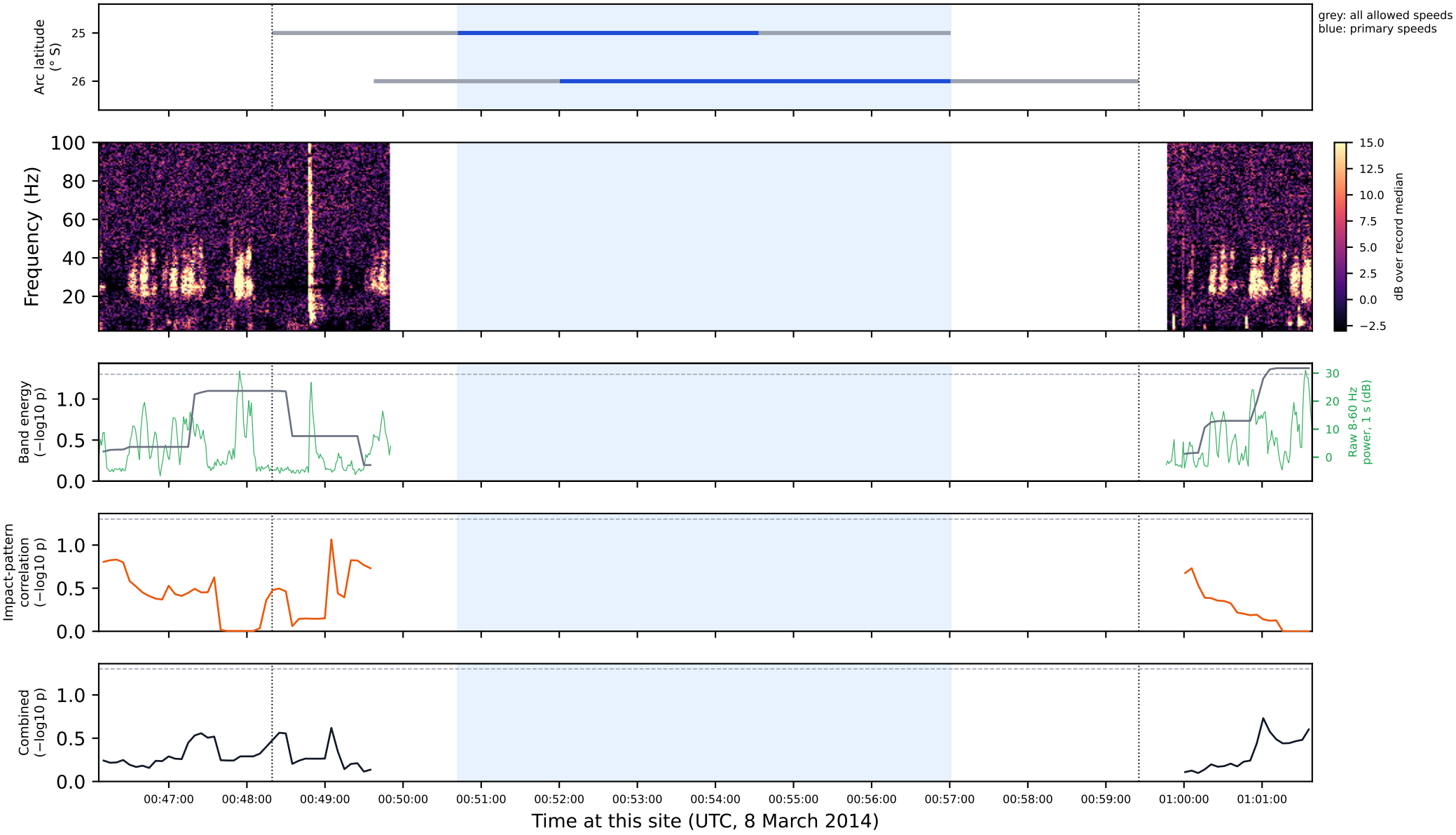
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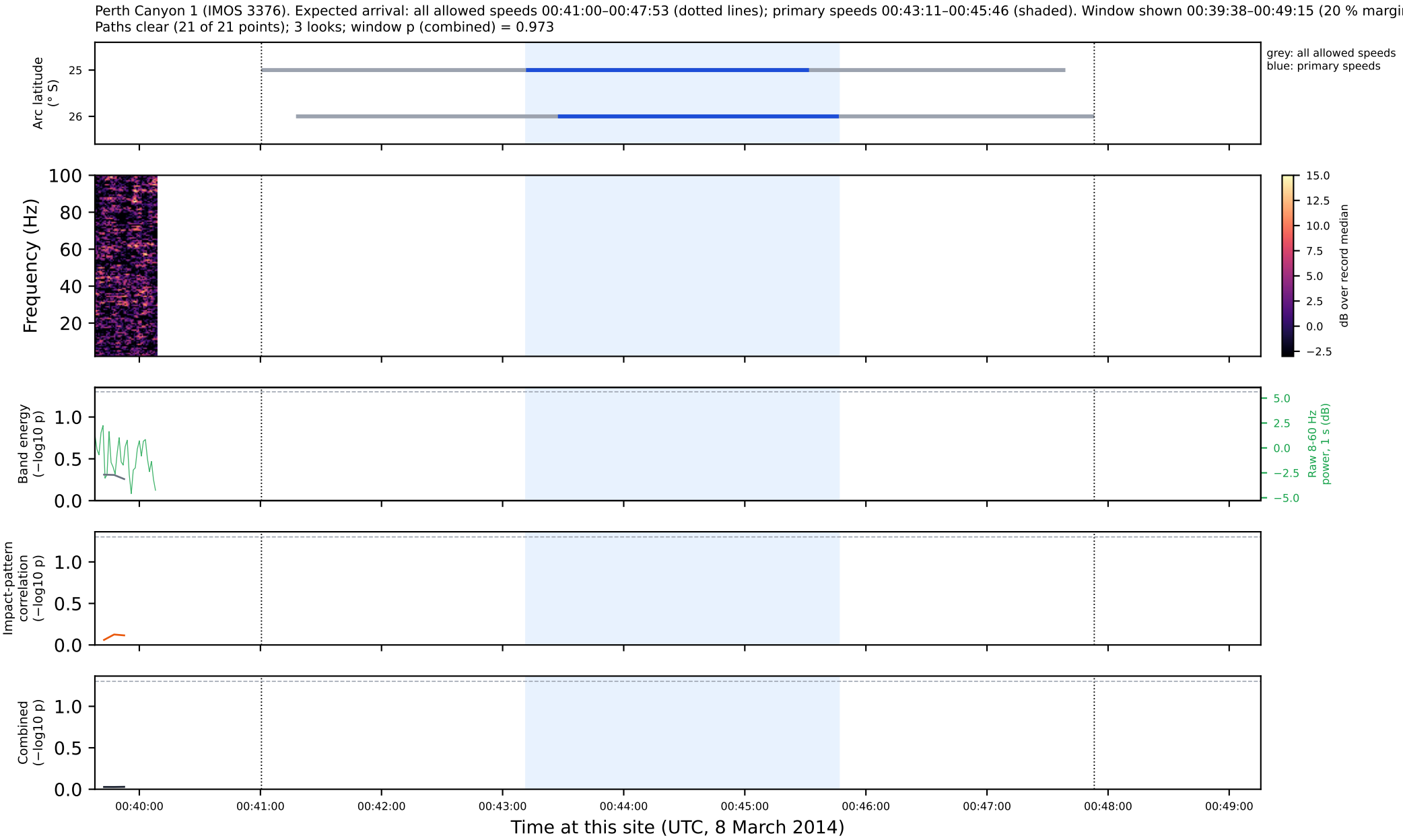
Candidate 8: Cape Leeuwin 00:45:36 (Kadri; bearing 306.1° corrected, 312.0° published). Implied impact 00:22:04–00:31:34 UTC, arc 25; 26° S

Scott Reef (IMOS 3250). Expected arrival: all allowed speeds 00:48:19–00:59:25 (dotted lines); primary speeds 00:50:42–00:57:00 (shaded). Window shown 00:46:06–01:01:38 (20 % margin).
Paths large attenuation (20 of 21 points); 62 looks; window p (combined) = 0.694



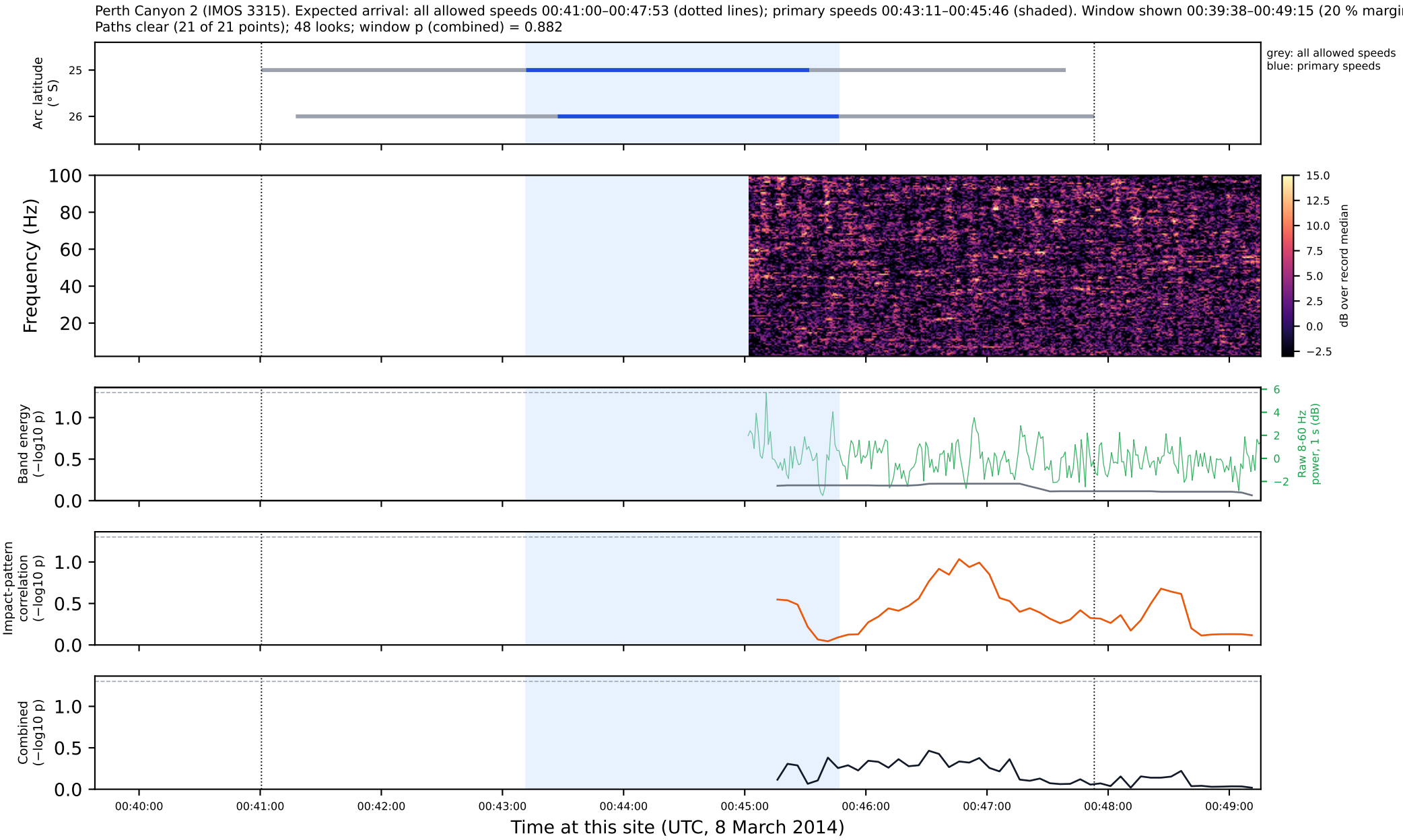
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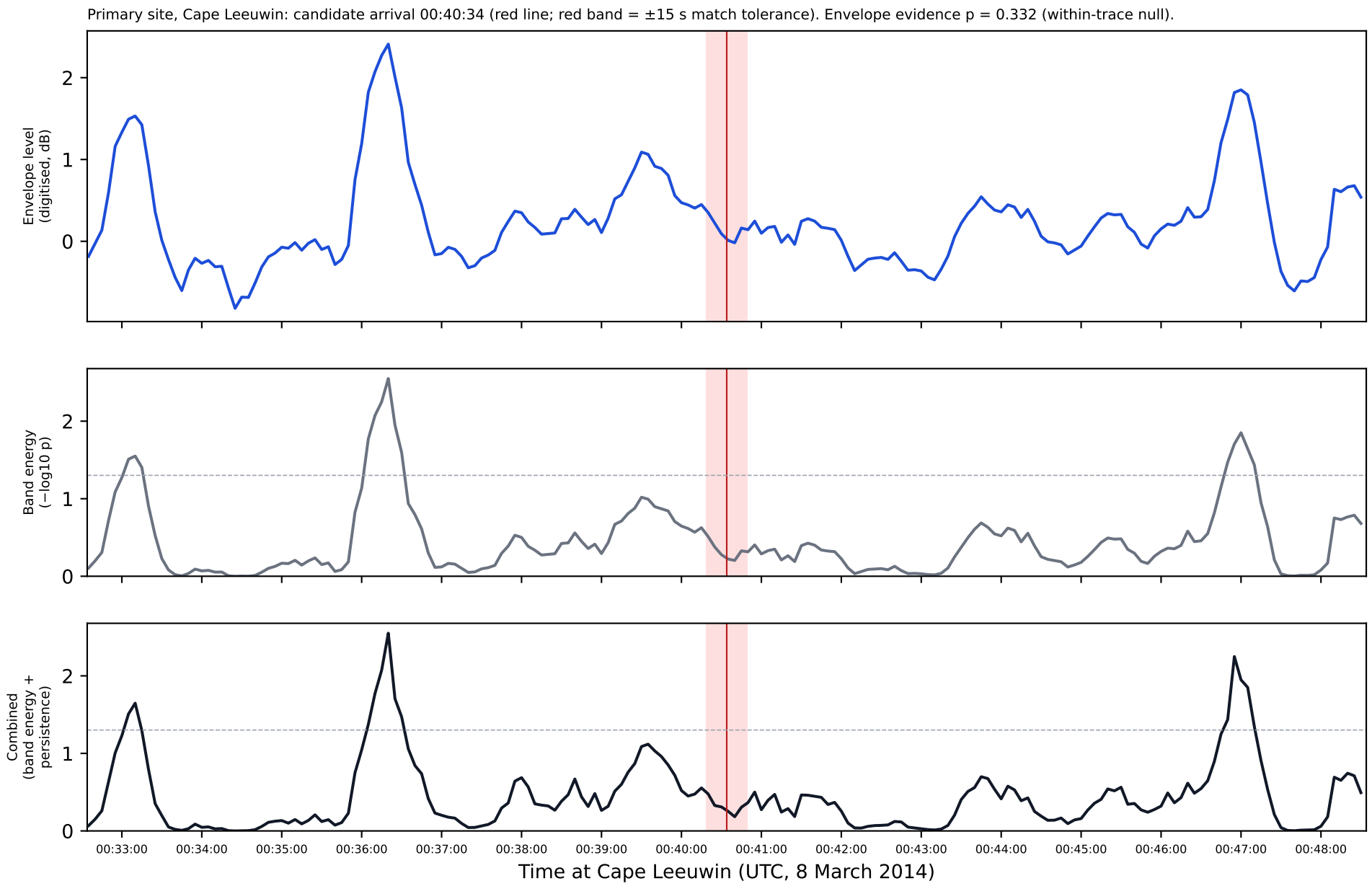
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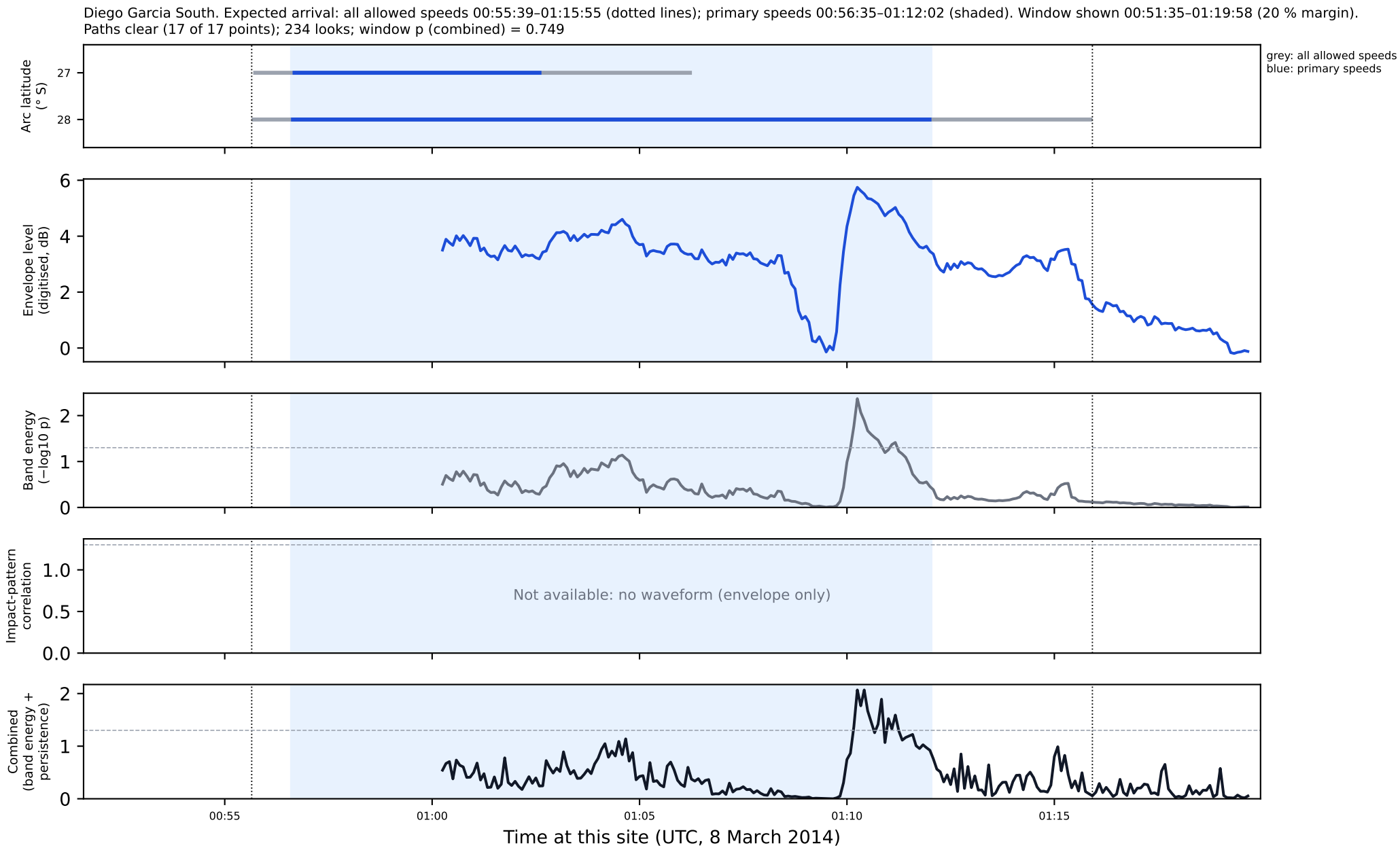
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Candidate 9: Cape Leeuwin 00:40:34 (Kadri; bearing 296.0° corrected, 303.3° published). Implied impact 00:19:38–00:26:50 UTC, arc 28° S



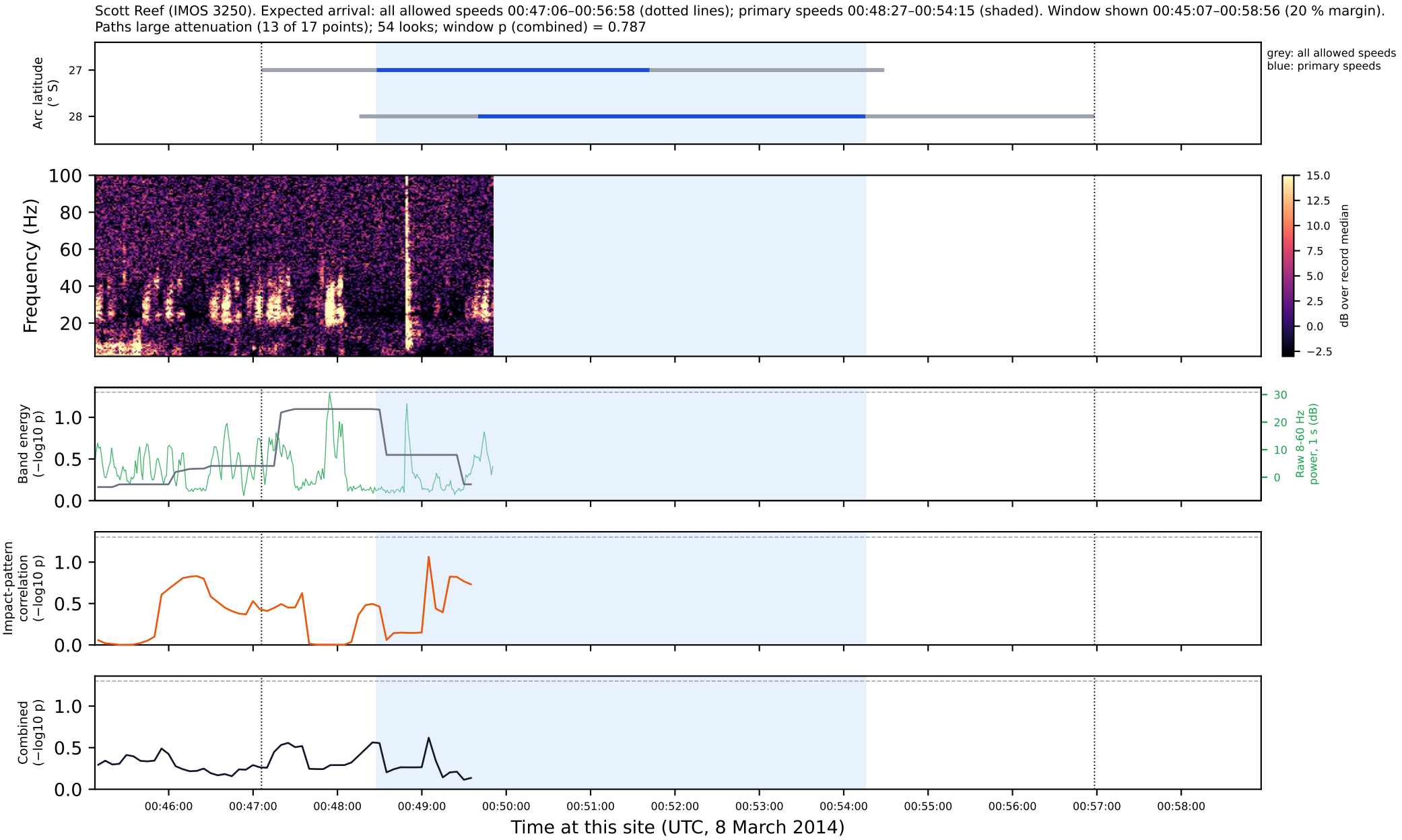
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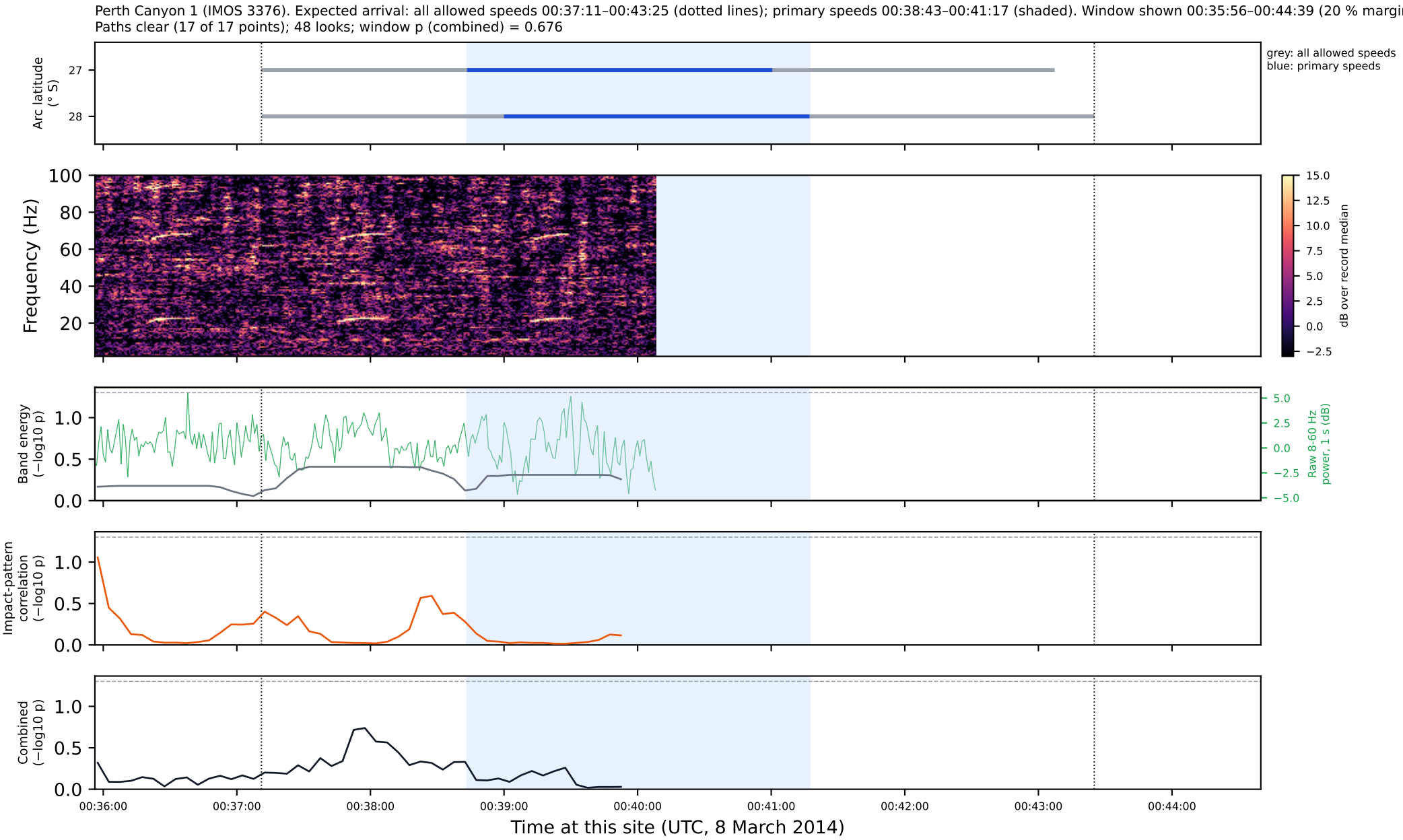
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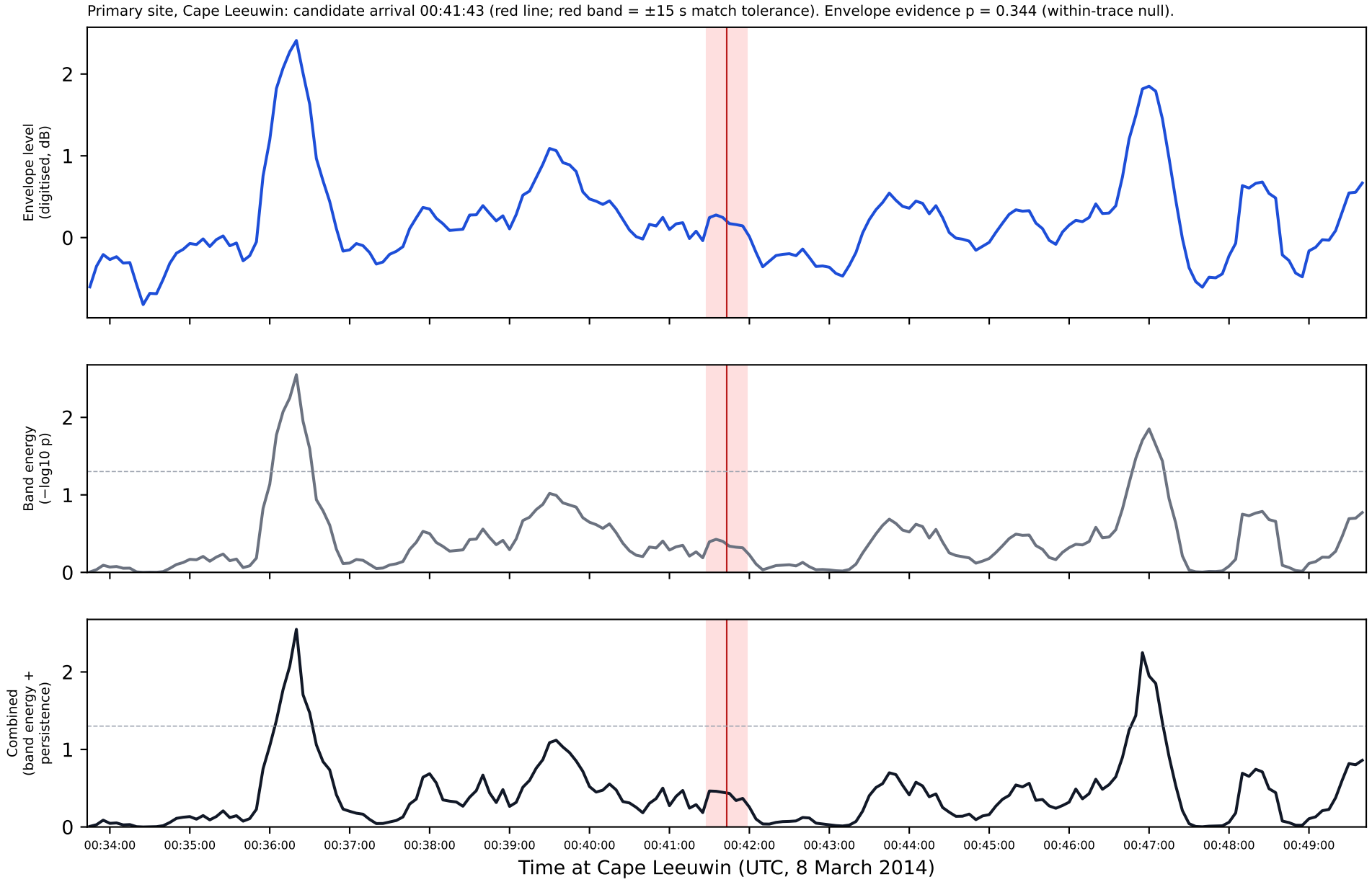
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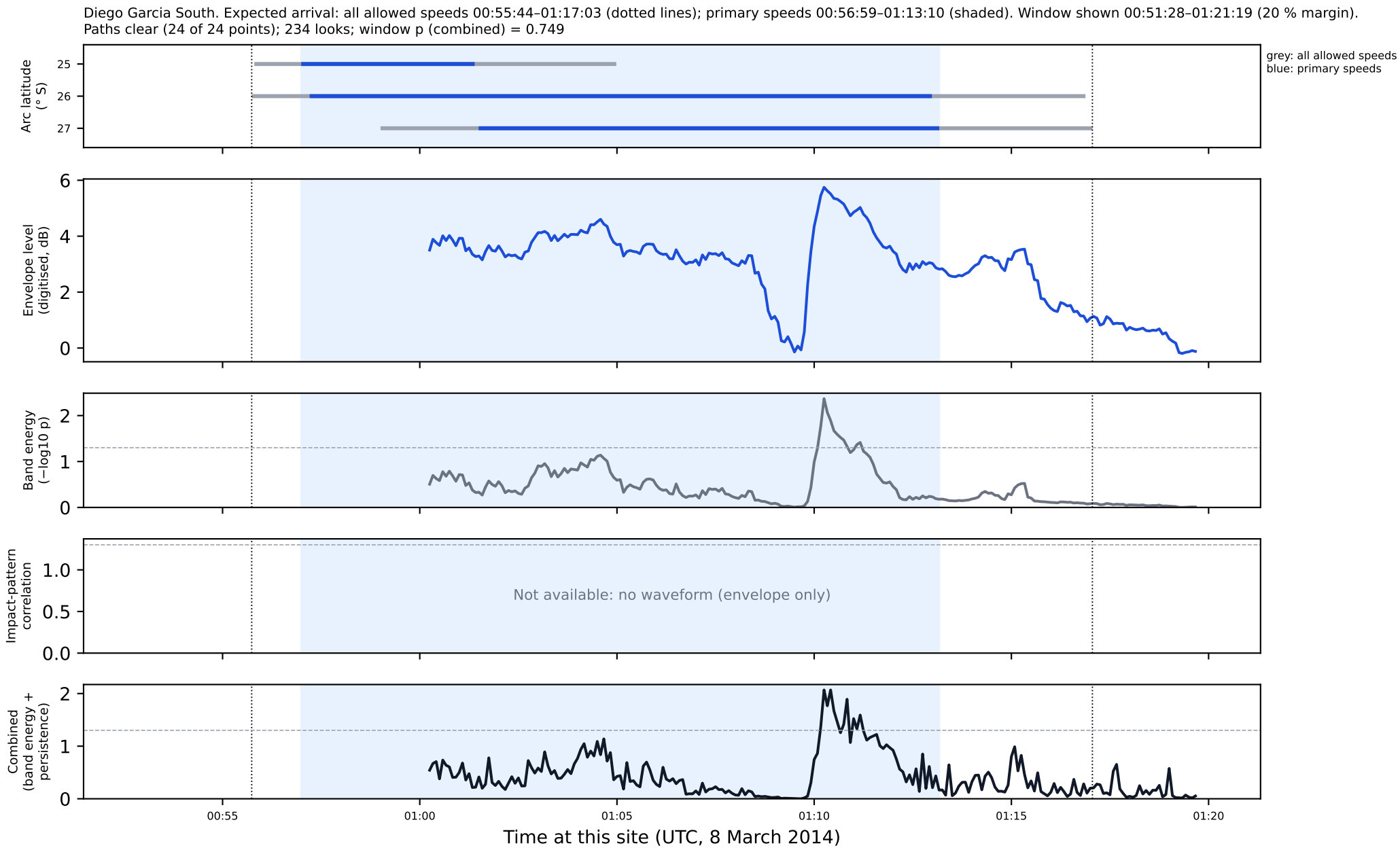
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Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:19:38–00:27:55 UTC, arc 26; 27° S



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is available for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). The p values are within-trace and not corrected for the number of looks; the p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05.

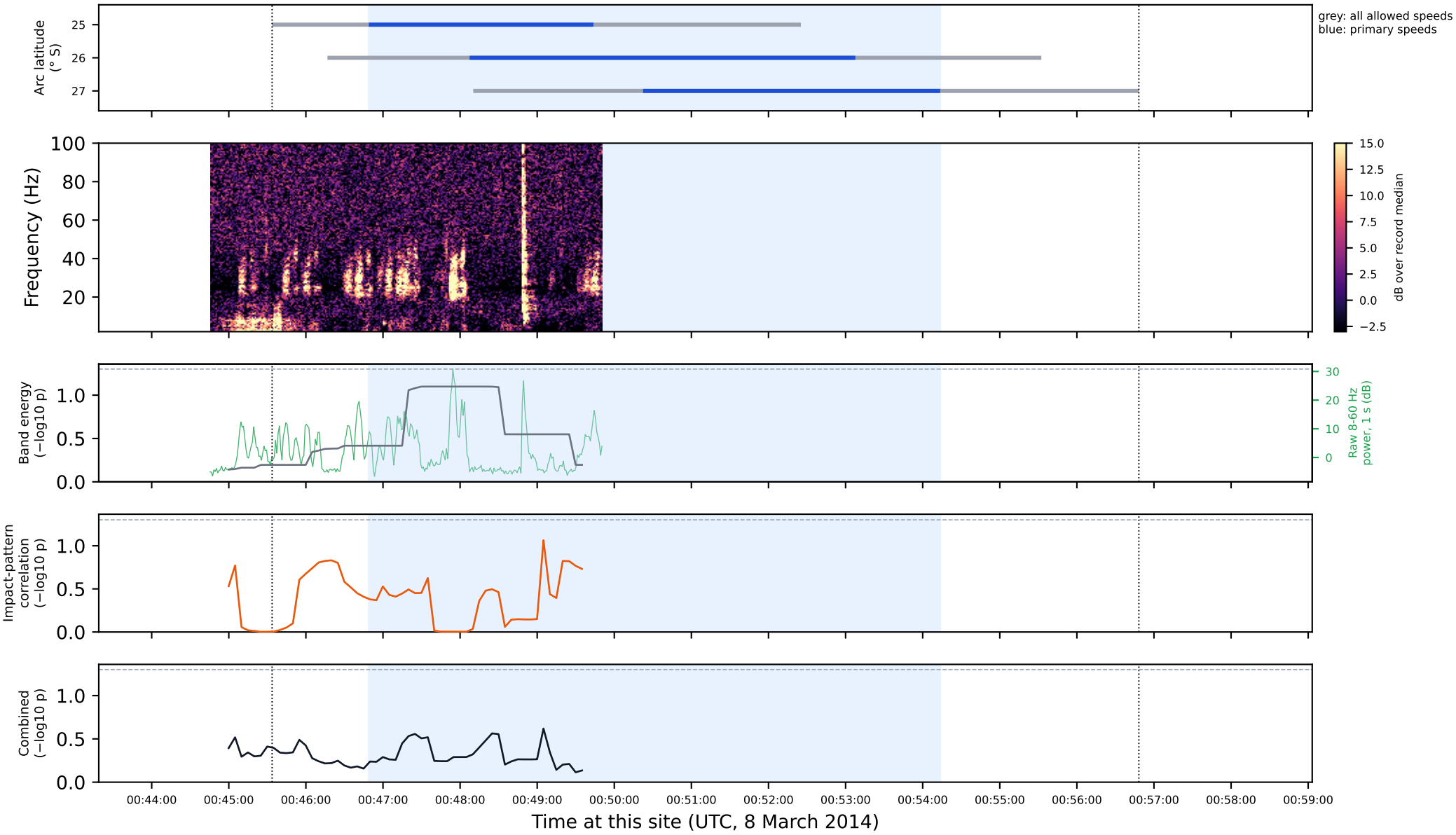
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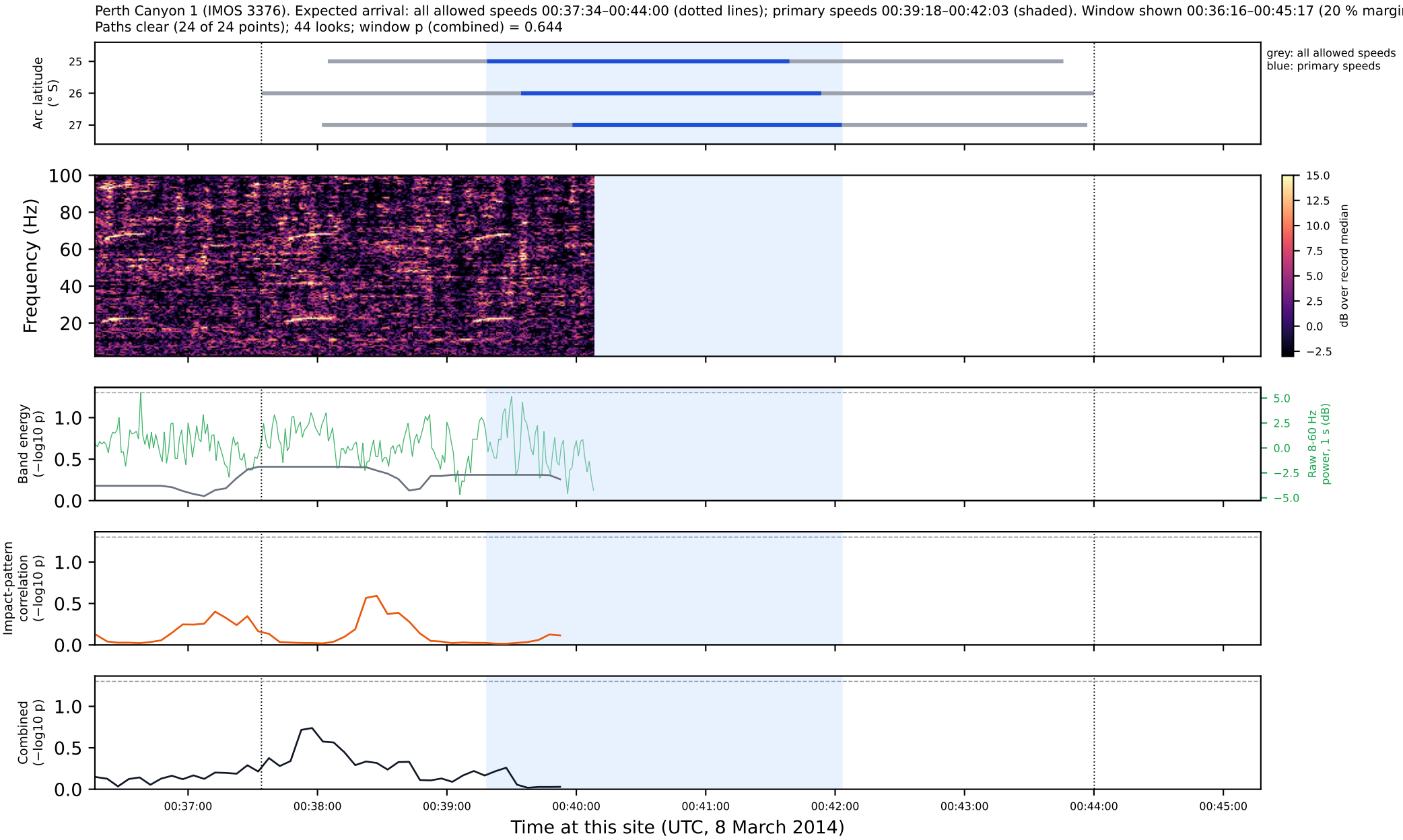
Candidate 10: Cape Leeuwin 00:41:43 (Kadri; bearing 303.8° corrected, 310.0° published). Implied impact 00:19:38–00:27:55 UTC, arc 26; 27° S

Scott Reef (IMOS 3250). Expected arrival: all allowed speeds 00:45:33–00:56:48 (dotted lines); primary speeds 00:46:49–00:54:13 (shaded). Window shown 00:43:18–00:59:03 (20 % margin).
Paths large attenuation (20 of 24 points); 56 looks; window p (combined) = 0.79



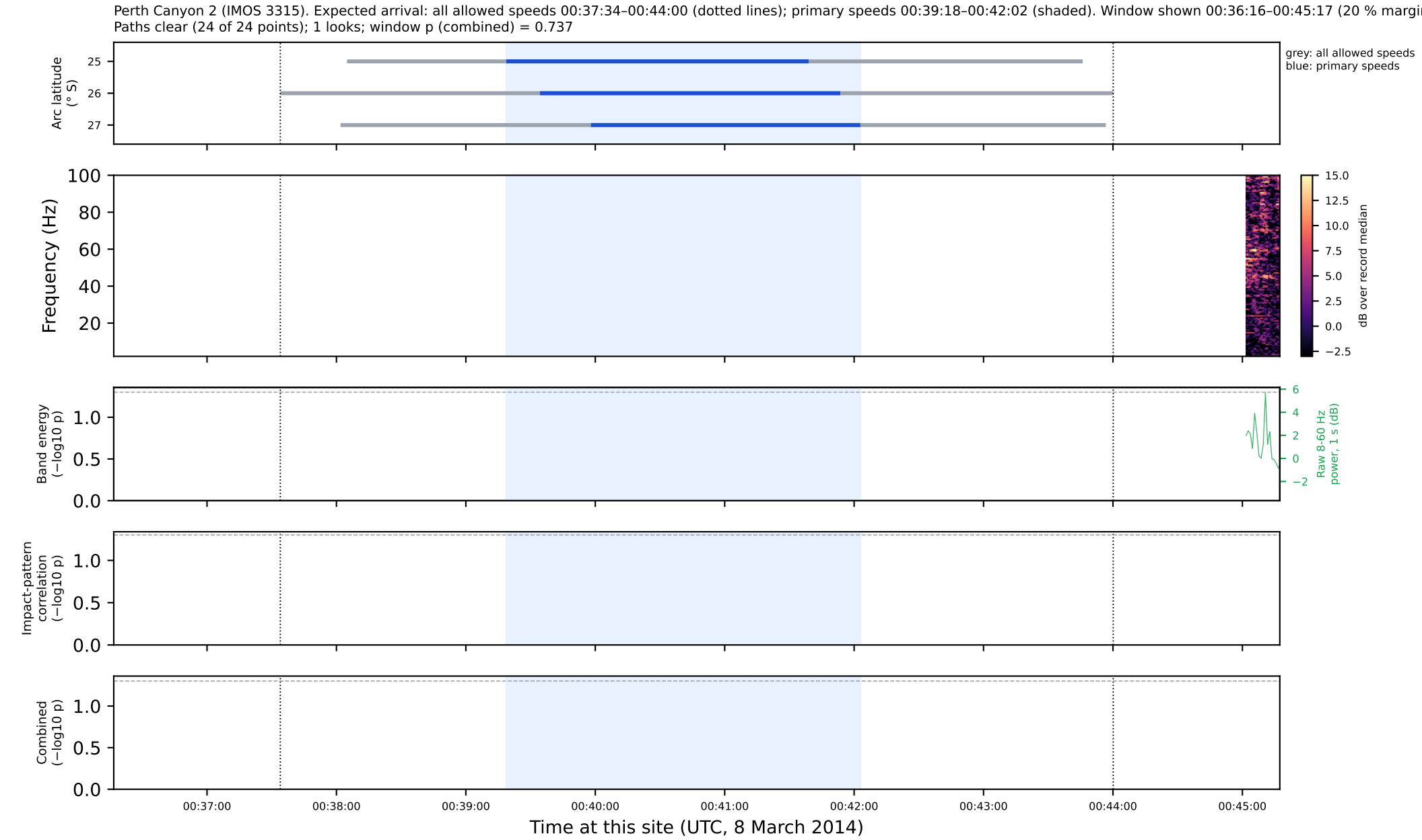
Window: earliest arrival at 1.56 km/s to latest at 1.40 km/s with 5 % path detour, over every impact point and time consistent with the candidate (corrected bearing within its tolerance when one is published; impact 00:19:38 to 00:55:00 UTC; 7th arc 15° S to 43° S within 150 NM, 20° S to 41° S when there is no bearing), then widened by 20 % of its length at each end (outer dotted lines mark the unwidened span). Shaded: span for the primary sound speed 1.465 to 1.520 km/s with 2 % detour. Sites shown: every site with an unblocked path (GEBCO 2026 screen) from at least one consistent point. IMOS loggers record 307 s in every 15 min; grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), from 14 days of noise (1 to 15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15 to 02:00; band energy 8 to 40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of all five scores. Noise calibration with USGS M≥3.5 earthquake arrival windows and faulty records removed (14-19 % of records). Window p: chance that noise alone gives a maximum this high over the looks in the window (calibrated for the look count). Cape Leeuwin and Diego Garcia South: only the digitised Kadri (2024) Figure 9 envelopes (no waveform). Top strip: expected arrival time at this site for each arc latitude that fits the candidate. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median), drawn so the time alignment can be checked. Each score uses 30 s windows and searches a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide centred on it. Calibration: earthquake-cleaned noise records, fault records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum).

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